

# Bell-Nonlocal Quantum Entanglement as Boundary-Level Compatibility

A Kernel-Role Determinacy Theorem for Bell-Witness Fixation, No-Signaling, and  
Tensor-Level Compatibility

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## Abstract

This manuscript proves a fixed-domain kernel-role determinacy theorem for finite Bell-nonlocal, no-signaling witness objects and then specializes it to Bell-certified finite-dimensional quantum entanglement. The kernel roles are not an optional overlay on physics: once one operational family is used as the same non-degenerate witness across settings, lawful transformations, retained records, and later classification, governed sameness already requires admissibility, standing, reference, and irreversibility. Target instantiation therefore carries governance occupancy; a determinate witness cannot exist for review while occupying no admissible governance class.

The formal architecture separates primitive candidacy from the normal form it ultimately forces. Finite operational Bell objects carry target, channel, wing-indexed standing-token, continuation, report/reuse, comparison, and act-time data. Object-indexed query sorts and dependent response families are explicit. Exactly recoverable simulations induce their effect transports from carrier maps, stochastic pushforward, recovery, and natural query reindexing; the transports are not independent extra isomorphisms. A lawful groupoid acts on the resulting effect functors. Candidate admissibility requires fixed-target reference, exact recovery of the admitted empirical witness after lawful Bell transport, a predeclared active interface, evaluated positive and negative standing obligations, support-disciplined realization, mandatory continuation generation, a final status class, and mandatory supported use. Standing-flow, repair, and hidden metric authority are excluded separately for kernel-licensed same-act transformations; they are not inserted as a circular clause of construction admissibility. It does not presuppose tensor compatibility, Bell-locality status, a causal model, or a transport form. Exact equality on inactive bookkeeping coordinates is neither required nor governance-bearing.

The Bell carrier is a compatible empirical model on a finite measurement cover. Empirical lawful equivalence is defined independently through exactly recoverable empirical simulations that induce affine bijections of global-assignment simplices commuting with all context marginals. Hence global-section status and Bell nonlocality are invariant. The context measures are defined before compatibility typing and recover the complete operational table. Raw compatibility data are specified by context measures, standing values, marginal maps, wing-local and joint realization support, continuation, automorphism covariance, comparison behavior, and report/reuse behavior.

A certificate is proved before the proof-bearing object  $T_B(P)$  is defined. Standing, continuation, and report/reuse effects are separate projections.

The central Bell-specific theorem is not uniqueness among objects already defined to equal the target, and its occupancy proof does not quantify over arbitrary inadmissible presentations. It states that every same-target candidate-admissible governance construction is forced, component by component, into the certified boundary-compatible standing form on the complete target-active practice. Coordinate-deviation exhaustion is separated from the later componentwise closure. Positive theorems fix active anchor, standing content, realization/continuation, final-status class, mandatory report/reuse behavior, and domain identity. Exact target-bearing realization follows; extensional governance rigidity and the singleton target-induced governance quotient are corollaries. Inactive bookkeeping differences remain skin rather than false counterexamples.

Operational marginal invariance excludes controllable communication, while the wing-indexed standing-flow theorem excludes same-domain standing creation, amplification, cross-wing transfer, metric admission authority, and same-act repair. In the finite-dimensional bipartite tensor-product and local-POVM carrier, every separable realization is Bell-local; Bell nonlocality therefore forces every realizing state in that carrier to be entangled. Post-quantum Bell-nonlocal no-signaling models remain within the general witness theorem but outside the quantum corollary.

The accompanying Lean 4 development directly verifies the finite denominator-32 vector/Born carrier, structural no-signaling, CHSH numerator 80, all sixteen deterministic local extrema, and finite Bell-local exclusion. Its endpoint bridge is conditional on declared operational-to-AASC structures. The weaker candidate-admissibility theorem, empirical-lawful groupoid, wing-indexed standing-flow predicates, transported Bell profile, coordinate-deviation exhaustion, completed role collapse, and target-active componentwise normal-form derivation are manuscript-level claims outside the released Lean layer.

**Main result.** Let a finite Bell-nonlocal, no-signaling empirical model  $P$  be instantiated as one non-degenerate reusable witness on a fixed Bell domain. Let  $C$  be any canonical Bell-governance construction that claims that same target and satisfies the primitive candidate-admissibility conditions of theorem 3.14; those conditions do not require its standing, continuation, or reuse effects to equal the canonical target effects. Then

$$C \equiv_{\text{gov}} C_P^*,$$

where

$$C_P^* = (O_P, d_P^*, \alpha_P^*, \text{stEff}_P(T_B(P)), \text{ctEff}_P(T_B(P)), \tau_P^*, \text{useEff}_P(T_B(P)), s_0).$$

Equivalently, candidate admissibility forces exact target-bearing realization and the boundary-compatible standing form. Target occupancy and derived rigidity therefore imply

$$\mathcal{Q}_P = \{[C]_{\equiv_{\text{gov}}} : \text{Claims}_{\mathcal{B}}(C, P) \wedge \text{Adm}_{\mathcal{B}}^{\text{cand}}(C; P)\}$$

has exactly one element. In the finite-dimensional bipartite tensor-product and local-POVM carrier, Bell nonlocality forces every realizing state to be entangled. The direct finite Lean certificate is `vectorBorn_flagship_spine`; the released deepest bundle is `reasonableDeepestFiniteEntanglement_spine`.

**Proof-class posture.** AASC is used as a constraint formalism over non-degenerate determinacy. Its object is not a new Bell dynamics but the identity and admissibility structure required when one operational family is reused as the same standing witness. Constructive dynamics, causal models, hidden-variable completions, process formalisms, and operational reconstructions remain legitimate at their declared scope. A correctness-relevant objection must break a named link: target-instantiation occupancy, vocabulary-neutral adequacy to kernel necessity, lawful groupoid construction, semantic effect factorization, an exact imported theorem or its Bell hypothesis, Bell-local/global-section correspondence, the typed construction of  $T_B(P)$ , compatibility equivalence, the typed-interface governance representation theorem, equivariance, a componentwise uniqueness result, standing-flow exclusion, or the finite-dimensional quantum specialization.

**Lean 4 companion boundary.** Release v0.8.0 of the Lean companion and its archival record, DOI 10.5281/zenodo.21287887, provide the finite mechanized support cited here [24]. The flagship module is `MaleyLean/Papers/QuantumEntanglementBoundaryCompatibility/DerivedDynamics.lean`; its direct witness certificate is `vectorBorn_flagship_spine`, and its deepest named bundle is `reasonableDeepestFiniteEntanglement_spine`. The released package covers the finite carrier and conditional endpoint routing. Target occupancy, the typed semantic interface, localized lawful-equivalence groupoid, compatibility equivalence, equivariant typed-interface representation, active

orbit-valued normalization, coordinate exhaustion, componentwise target-active uniqueness, and quantum specialization are manuscript-level results and are not attributed to the released code.

**Paired quantum determinacy arc.** This manuscript is the Bell-witness member of the paired AASC quantum determinacy arc. The companion measurement-record theorem establishes that physical record occurrence becomes non-degenerate evidential record objecthood only through admissible record-role fixation. The present paper establishes that physical Bell correlation occurrence becomes non-degenerate reusable witness objecthood only through a fixed compatibility structure. The record theorem is paired support and handoff context, not a premise of the local Bell proof.

**Kernel theorem and kernel roles.** The dependency is

$$\begin{aligned} \text{non-degenerate same-witness use} &\implies \text{governed sameness and target occupancy} \\ &\implies \{\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}\}. \end{aligned}$$

Reference fixes which operational family is the same witness. Standing fixes its eligibility for theorem-bearing and evidential reuse. Admissibility separates lawful preservation from selector import, hidden load, and target change. Irreversibility prevents later calibration or selection from rewriting the same earlier act. A practice may satisfy all four tacitly. Vocabulary is optional; the roles are not. An alleged determinate witness with no admissible governance realization is not an empty case; it is a refusal or retyping of the antecedent target.

**Proof spine.** The live proof order is acyclic:

- (1) Operational Bell objects, explicit object-indexed query sorts, dependent response families, and exactly recoverable simulations are defined. Their effect isomorphisms extend to six functors on the lawful groupoid  $\mathcal{G}_{\mathcal{B}}$ .
- (2) The fixed Bell domain, raw typed semantic presentations, syntactic six-coordinate restriction, extensional governance equivalence, canonical tuple carrier, and orbit-valued normalization are defined without invoking semantic exhaustion.
- (3) Instantiated non-degenerate witness use yields governed sameness, target occupancy, and the Bell kernel.
- (4) Only then do the imported fixed-domain theorems establish no second classifier, semantic governance-locus exhaustion, unique effect-channel routing, standing conservation, AMetric boundary closure, trace fixation, and singleton-interior rigidity.
- (5) The operational family defines a compatible empirical model, exact context realization family, Bell-local completion space, and global-section obstruction.

- (6) Raw compatibility data are constructed, certified, and promoted to  $T_B(P)$ . Standing, continuation, and report/reuse effects remain separate; compatibility equivalence is equality of their full transported profile.
- (7) Primitive candidate admissibility, kernel-licensed standing-flow exclusion, exact-faithful compatibility, the exact-standing quotient interface, and coordinate-deviation exhaustion are proved before positive target-active componentwise uniqueness. Full role collapse is then a corollary of the completed componentwise closure.
- (8) The boundary-compatible standing-form theorem derives exact canonical behavior from weaker candidate admissibility. Occupancy plus derived rigidity gives the singleton governance quotient as a corollary.
- (9) Operational no-signaling and the explicit standing-flow theorem yield the no-signaling determinacy endpoint; finite-dimensional quantum realizability supplies the Bell-entanglement specialization.

**Finite mechanized companion spine.** The Lean companion directly verifies the finite Bell carrier: scaled integer vector geometry, vector/Born weights, structural no-signaling, CHSH numerator 80, exhaustive deterministic local-response classification, denominator-32 Bell-local-envelope exclusion, and a declared conditional bridge to the boundary/no-signaling surface. It is support for the finite carrier and formal routing, not an additional premise of the manuscript-level uniqueness theorem.

## Contents

|          |                                                                               |           |
|----------|-------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                                           | <b>7</b>  |
| <b>2</b> | <b>Operational Bell objects, lawful localization, and typed governance</b>    | <b>9</b>  |
| 2.1      | Operational objects and object-indexed query carriers . . . . .               | 9         |
| 2.2      | Lawful equivalence and its effect functors . . . . .                          | 12        |
| 2.3      | Fixed domain and typed semantic presentations . . . . .                       | 13        |
| 2.4      | Physical occurrence and the neutral target . . . . .                          | 19        |
| 2.5      | Boundary traces for Bell witness channels . . . . .                           | 20        |
| <b>3</b> | <b>Kernel activation, semantic exhaustiveness, and primitive conservation</b> | <b>21</b> |
| 3.1      | Bell standing-flow predicates . . . . .                                       | 27        |

|           |                                                                            |           |
|-----------|----------------------------------------------------------------------------|-----------|
| <b>4</b>  | <b>Bell empirical models, obstruction packet, and tensor compatibility</b> | <b>30</b> |
| 4.1       | Empirical model and standing quotient . . . . .                            | 30        |
| 4.2       | Bell-local completion and global-section obstruction . . . . .             | 32        |
| 4.3       | Exact context realization before compatibility typing . . . . .            | 33        |
| 4.4       | Bell joint obstruction packet . . . . .                                    | 34        |
| 4.5       | Raw compatibility data, certification, and exact equivalence . . . . .     | 35        |
| <b>5</b>  | <b>AMetric co-standing and exact local realization</b>                     | <b>42</b> |
| <b>6</b>  | <b>Measurement as admissible local realization</b>                         | <b>43</b> |
| <b>7</b>  | <b>Bell-local completion, hidden variables, and richer-scope models</b>    | <b>44</b> |
| <b>8</b>  | <b>No-signaling hierarchy and same-domain transport exclusion</b>          | <b>45</b> |
| <b>9</b>  | <b>Metric representation is downstream of Bell-witness fixation</b>        | <b>47</b> |
| <b>10</b> | <b>CHSH audit and Bell-governance operators</b>                            | <b>48</b> |
| 10.1      | Input and output presentations . . . . .                                   | 48        |
| 10.2      | Finite Lean mechanization of the Bell-side witness . . . . .               | 48        |
| 10.3      | Semantic standing-relevant operators . . . . .                             | 49        |
| 10.4      | Canonical target representative and positive uniqueness . . . . .          | 50        |
| <b>11</b> | <b>Bell-witness theorem and finite-dimensional quantum specialization</b>  | <b>53</b> |
| <b>12</b> | <b>Mechanism constraint and paired handoff</b>                             | <b>55</b> |
| <b>13</b> | <b>Scope, nonclaims, and denial burden</b>                                 | <b>56</b> |
| 13.1      | What the theorem establishes . . . . .                                     | 56        |
| 13.2      | What the theorem does not claim . . . . .                                  | 57        |
| 13.3      | Standing-filtered denial matrix . . . . .                                  | 57        |
| <b>14</b> | <b>Conclusion</b>                                                          | <b>58</b> |
| <b>A</b>  | <b>Paired quantum determinacy ledger</b>                                   | <b>60</b> |
| <b>B</b>  | <b>Bell-scope and mechanization certificate</b>                            | <b>60</b> |

|          |                                                        |           |
|----------|--------------------------------------------------------|-----------|
| <b>C</b> | <b>Boundary and non-claim guide</b>                    | <b>61</b> |
| <b>D</b> | <b>Co-standing glossary</b>                            | <b>61</b> |
| <b>E</b> | <b>Theorem-status map</b>                              | <b>62</b> |
| <b>F</b> | <b>Lean 4 mechanization ledger</b>                     | <b>63</b> |
| <b>G</b> | <b>Imported theorem and Bell-specialization ledger</b> | <b>63</b> |
| <b>H</b> | <b>Reference integrity and publication notes</b>       | <b>66</b> |

## 1 Introduction

Quantum entanglement presents a familiar structural split. Bell and CHSH show that suitable operational correlations cannot be reproduced by Bell-local hidden-variable factorization, while the same operational tables satisfy no-signaling constraints and cannot be used for controllable superluminal communication [2, 3, 4]. The present manuscript asks a different but prior question: what must be fixed when that table is used as the same non-degenerate Bell witness across settings, lawful relabelings, retained records, and later classification?

The theorem does not choose a mechanism generating the correlations. It identifies the standing-bearing form of the reusable witness. Physical occurrence supplies preparations, settings, detector events, outcomes, records, and estimated probabilities. Bell-witness determinacy adds governed sameness: one fixed operational family, one declared interface package, one lawful transformation class, one act-time status, and one licensed reuse profile. Once that target is instantiated, the kernel roles are identity conditions rather than optional terminology, and the target already occupies an admissible governance form. The proof then concerns rigidity: no second governance-non-equivalent form may faithfully realize the same determinate object.

The mathematical carrier is a finite Bell empirical model: a family of context distributions on the measurement cover whose restrictions agree on overlaps. Bell-local completion is the existence of a global joint distribution or, equivalently in the finite setting, a convex mixture of deterministic global assignments. Fine established the CHSH/global-joint-distribution equivalence in the binary scenario, while the sheaf-theoretic framework identifies Bell nonlocality as obstruction to a global section on the measurement cover [5, 8]. This manuscript uses that established mathematics to construct a Bell joint obstruction packet and an exact realization family before defining the compatibility object.

The locality vocabulary is kept disciplined. Jarrett and Shimony distinguish locality conditions often discussed as parameter independence and outcome independence [6, 7]. The present no-

signaling hierarchy does not replace that analysis: it separates operational marginal invariance, absence of a controllable communication channel, and exclusion of same-domain standing-transport typing. Operational-equivalence and ontological-model analyses likewise show why identity of procedures and representations matters when moving from observed statistics to underlying models [9]. Causal-model work demonstrates that no-signaling Bell violations impose nontrivial constraints on causal explanations and may require fine-tuning under standard causal assumptions [10, 11]. Those richer-scope formalisms are not excluded. The semantic representation theorem classifies only the governance effect they claim on the fixed witness: anchor admission, standing content, admissible continuation, act-time status, target identity, or no independent effect.

No-signaling correlations extend beyond the quantum set. Popescu–Rohrlich and device-independent analyses exhibit Bell-nonlocal no-signaling models that are not quantum-realizable [12, 13]. Conversely, entanglement alone does not guarantee Bell nonlocality; Werner states supply the standard separation [14]. Accordingly, the strongest theorem below is first stated for finite Bell-nonlocal no-signaling empirical models. The quantum specialization adds a Hilbert-space realization; separable realizations are Bell-local, so Bell violation forces the realizing state to be entangled within the specified carrier.

The released Lean companion mechanizes a concrete finite carrier. It derives an exact denominator-32 vector/Born table from integer geometry, proves normalization and structural no-signaling, obtains CHSH numerator 80, classifies all sixteen deterministic local assignments, and excludes the witness from the denominator-32 Bell-local envelope. The manuscript supplies the distinct structural layer: target occupancy, semantic governance representation, extensional governance equivalence, the Bell joint obstruction packet, exact compatibility certification, positive target-active componentwise uniqueness, and the separation between the general Bell-nonlocal result and the quantum-entanglement corollary.

**Positive Bell-side interpretation.** The numerical values come from the operational or quantum carrier. AASC supplies the admissible same-domain type of the object carrying them. The local wings are not treated as independently complete standing states later joined by a message. They are local roles of one compatible empirical model, fixed through declared interfaces and sampled context by context. “AMetric co-standing” names this admission-level relation; it is not a hidden surface, an influence, or a derivation of the probability values.

**Handoff from the measurement-record theorem.** The present paper governs the standing structure of the Bell witness sampled by local records. The companion paper governs whether those records can be reused as the same evidential objects. The two results are coordinated but logically distinct.

## 2 Operational Bell objects, lawful localization, and typed governance

### 2.1 Operational objects and object-indexed query carriers

**Definition 2.1** (Finite operational Bell object). Fix ambient sets of target identifiers, act identifiers, status values, comparison tests, report tokens, and finite presentation labels, together with an act-support map

$$\ell^{\text{act}} : \text{Act} \rightarrow \{A, B, AB\}.$$

A finite operational Bell object is a tuple

$$O = (X_O, Y_O, \{A_x^O\}, \{B_y^O\}, P_O, \mathcal{C}_O, \ell_O^C, \alpha_O, \theta_O, Z_O, \sigma_O, \ell_O^Z, \eta_O, S_O, \ell_O^S, \mathcal{K}_O, \text{Rpt}_O, \ell_O^R, \rho_O, \zeta_O, W_O, \omega_O, \tau_O),$$

where  $X_O, Y_O$  and all outcome sets are finite;  $P_O(a, b \mid x, y)$  is normalized;  $\mathcal{C}_O$  is the declared channel/interface carrier with wing map  $\ell_O^C : \mathcal{C}_O \rightarrow \{A, B, AB\}$ ;  $\alpha_O : \mathcal{C}_O \rightarrow \{0, 1\}$  is its reference anchor-admission verdict;  $\theta_O$  is the target identifier;  $Z_O$  is the finite set of evaluated witness tokens with reference standing valuation  $\sigma_O : Z_O \rightarrow \{0, 1\}$  and wing label

$$\ell_O^Z : Z_O \rightarrow \{A, B, AB\};$$

$\eta_O$  is the witness-act identifier;  $S_O$  is the reuse-state carrier with wing label  $\ell_O^S : S_O \rightarrow \{A, B, AB\}$  and reference continuation relation  $\mathcal{K}_O \subseteq S_O \times S_O$ ;  $\text{Rpt}_O$  is the report-token carrier with wing label  $\ell_O^R : \text{Rpt}_O \rightarrow \{A, B, AB\}$ , reference report/reuse verdict  $\rho_O : \text{Rpt}_O \sqcup S_O \rightarrow \{0, 1\}$ , and support map  $\zeta_O : \text{Rpt}_O \sqcup S_O \rightarrow Z_O$ ;  $W_O$  is a finite family of functions from empirical tables to  $\{-1, 0, 1\}$ , with reference comparison profile  $\omega_O(w) = w(P_O)$ ; and  $\tau_O$  is the reference act-time status. Write  $\mathcal{C}_O^A, \mathcal{C}_O^B, \mathcal{C}_O^{AB}, Z_O^A, Z_O^B, Z_O^{AB}, S_O^A, S_O^B, S_O^{AB}$ , and  $\text{Rpt}_O^A, \text{Rpt}_O^B, \text{Rpt}_O^{AB}$  for the corresponding label fibers.

The displayed values determine the reference section formalized next. They specify the declared target packet used to type lawful transport; they are not a second candidate governance proposal. Candidate governance values are supplied only by the semantic answer section introduced below. No semantic claim that the six later governance loci are exhaustive is used here.

**Definition 2.2** (Reference effect section). For an operational object  $O$ , let

$$\text{RefSec}(O) := (d_O^{\text{ref}}, a_O^{\text{ref}}, t_O^{\text{ref}}, c_O^{\text{ref}}, \tau_O^{\text{ref}}, u_O^{\text{ref}})$$

be the six effect sections induced by the target identifier, cover, channel package, act identifier, anchor verdict  $\alpha_O$ , empirical table  $P_O$ , standing valuation  $\sigma_O$ , realization supports and continuation  $\mathcal{K}_O$ , comparison profile  $\omega_O$ , act-time status  $\tau_O$ , and report/reuse verdict  $\rho_O$ . Write  $\text{RefSec}_L(O)$  for its  $L$ -coordinate.

**Definition 2.3** (Object-indexed governance queries and dependent response families). For each operational object  $O$ , define six explicit query sorts:

$$\begin{aligned}
Q_D(O) &:= \{q_{\text{target}}, q_{\text{cover}}, q_{\text{channels}}, q_{\text{act}}\}, \\
Q_A(O) &:= \{q_{\text{admit}}(c) : c \in \mathcal{C}_O\}, \\
Q_T(O) &:= \{q_{\text{cell}}(x, y, a, b)\} \sqcup \{q_{\text{stand}}(z) : z \in \mathbf{Z}_O\} \sqcup \{q_{\text{cmp}}(w) : w \in \mathbf{W}_O\}, \\
Q_C(O) &:= \{q_{\text{real}}^{AB}(x, y, a, b)\} \sqcup \{q_{\text{real}}^A(x, a)\} \sqcup \{q_{\text{real}}^B(y, b)\} \sqcup \{q_{\text{cont}}(s, s') : s, s' \in \mathbf{S}_O\}, \\
Q_\tau(O) &:= \{q_{\text{status}}(\eta_O)\}, \\
Q_U(O) &:= \{q_{\text{report}}(r) : r \in \mathbf{Rpt}_O\} \sqcup \{q_{\text{reuse}}(s) : s \in \mathbf{S}_O\}.
\end{aligned}$$

The omitted ranges in  $q_{\text{cell}}$  and  $q_{\text{real}}^{AB}$  are  $x \in X_O$ ,  $y \in Y_O$ ,  $a \in A_x^O$ , and  $b \in B_y^O$ . Fix ambient typed carriers **Target**, **Cover**, **ChanPkg**, **Act**, and **Status**. Define the dependent response family by

|             |                     |                    |                       |                  |
|-------------|---------------------|--------------------|-----------------------|------------------|
| $q$         | $q_{\text{target}}$ | $q_{\text{cover}}$ | $q_{\text{channels}}$ | $q_{\text{act}}$ |
| $V_D(O, q)$ | <b>Target</b>       | <b>Cover</b>       | <b>ChanPkg</b>        | <b>Act</b>       |

and by

$$\begin{aligned}
V_A(O, q_{\text{admit}}(c)) &= \{0, 1\}, \\
V_T(O, q_{\text{cell}}) &= [0, 1], \quad V_T(O, q_{\text{stand}}) = \{0, 1\}, \quad V_T(O, q_{\text{cmp}}) = \{-1, 0, 1\}, \\
V_C(O, q) &= \{0, 1\}, \quad V_\tau(O, q) = \mathbf{Status}, \quad V_U(O, q) = \{0, 1\}.
\end{aligned}$$

Define the object-indexed effect section

$$\mathcal{X}_L(O) := \prod_{q \in Q_L(O)} V_L(O, q), \quad L \in \{D, A, T, C, \tau, U\},$$

and the aliases

$$\mathcal{B}_B(O) := \mathcal{X}_D(O), \quad \mathcal{A}_B(O) := \mathcal{X}_A(O), \quad \mathcal{T}_B(O) := \mathcal{X}_T(O), \quad \mathcal{C}_B(O) := \mathcal{X}_C(O),$$

$$\mathcal{T}_B^{\text{act}}(O) := \mathcal{X}_\tau(O), \quad \mathcal{U}_B(O) := \mathcal{X}_U(O).$$

Define the support query map  $\bar{\zeta}_O : Q_U(O) \rightarrow \mathbf{Z}_O$  by

$$\bar{\zeta}_O(q_{\text{report}}(r)) := \zeta_O(r), \quad \bar{\zeta}_O(q_{\text{reuse}}(s)) := \zeta_O(s).$$

The complete typed query carrier is the displayed disjoint union  $Q_B(O) := \bigsqcup_L Q_L(O)$ ; it is a syntactic interface, not yet a semantic exhaustion theorem.

**Definition 2.4** (Exactly recoverable operational simulation). An exactly recoverable simulation  $f : O \rightarrow O'$  consists of:

(S1) setting maps  $f_X, f_Y$ , contextwise stochastic kernels

$$K_x^{A,f} : A_x^O \rightarrow \Delta(A_{f_X x}^{O'}), \quad K_y^{B,f} : B_y^O \rightarrow \Delta(B_{f_Y y}^{O'}),$$

and recovery data of the same type;

(S2) maps on channel, standing-token, reuse-state, report, and comparison carriers preserving the channel, token, reuse-state, and report wing labels, support maps, target identifiers, and act identifiers on the admitted witness;

(S3) the stochastic pushforward and recovery equations

$$P_{O'} = f_* P_O, \quad P_O = r_{f*} P_{O'},$$

with both composites equal to the identity on the admitted witness table;

(S4) query reindexing maps  $f_L^\#$  and  $r_L^\#$  induced by the carrier maps for every non-cell query, together with dependent fiber maps

$$\varphi_{f,q}^L : \mathbf{V}_L(O, q) \longrightarrow \mathbf{V}_L(O', f_L^\# q)$$

that are inverse to the corresponding recovery maps on admitted query fibers;

(S5) induced section transports  $F_f^L$ , not independent extra data: on empirical-cell coordinates  $F_f^T$  is the stochastic pushforward by  $K^{A,f} \otimes K^{B,f}$ ; on standing, comparison, anchor, continuation, act-time, and report/reuse coordinates it is dependent reindexing through  $f_L^\#$  and  $\varphi_{f,q}^L$ ; and local realization coordinates are the marginals of the transported joint realization support;

(S6) reference-section naturality:

$$F_f^L(\text{RefSec}_L(O)) = \text{RefSec}_L(O') \quad (L \in \{D, A, T, C, \tau, U\}),$$

where  $\text{RefSec}_L$  is the  $L$ -coordinate of theorem 2.2.

The induced transports satisfy

$$F_f^L : \mathcal{X}_L(O) \xrightarrow{\cong} \mathcal{X}_L(O'), \quad R_f^L : \mathcal{X}_L(O') \xrightarrow{\cong} \mathcal{X}_L(O),$$

$$R_f^L F_f^L = \text{id}, \quad F_f^L R_f^L = \text{id},$$

and commute with the query and response maps. Globally section-isomorphic refinements, dilations, ancillas, coarse-grainings, convex re-expressions, or implementation changes enter the lawful class only when the complete section spaces are isomorphic, the reference section is natural, and the exact recovery equations are proved. Thus the lawful category contains globally section-isomorphic exactly recoverable simulations, not arbitrary operational coarse-grainings.

**Definition 2.5** (Base operational category). Let  $\mathbf{Op}_{\mathcal{B}}$  be the category whose objects are finite operational Bell objects and whose arrows are exactly recoverable operational simulations. Identity and composition are componentwise; effect transport satisfies

$$F_{g \circ f}^L = F_g^L \circ F_f^L, \quad F_{\text{id}}^L = \text{id}.$$

**Definition 2.6** (Canonical Bell transformation). A canonical Bell transformation is an exactly recoverable simulation whose setting, outcome, channel, token, reuse, report, and comparison maps are bijections and whose transport maps are the induced reindexings. Its probability action is

$$(g \cdot P)(\sigma_x^A(a), \sigma_y^B(b) \mid \sigma_X(x), \sigma_Y(y)) = P(a, b \mid x, y).$$

Write  $\mathcal{G}_{\mathcal{B}}^{\text{can}}$  for the resulting isomorphism groupoid inside  $\mathbf{Op}_{\mathcal{B}}$ .

**Lemma 2.7** (Canonical Bell transformations form a groupoid). *The identity belongs to  $\mathcal{G}_{\mathcal{B}}^{\text{can}}$ , canonical transformations compose, and each has a canonical inverse.*

*Proof.* All carrier maps are bijections and all effect transports are reindexing isomorphisms. Identity, composition, and inverse are componentwise.  $\square$

**Theorem 2.8** (Canonical-invariance theorem). *Canonical Bell transformations preserve normalization, overlap consistency, operational no-signaling, global-section status, comparison profile, anchor verdicts, act identity, continuation, and licensed report/reuse.*

*Proof.* Bijective pushforward preserves total probability and commutes with marginalization. The remaining statements are invariance of the corresponding typed effect sections under the reindexing isomorphisms.  $\square$

## 2.2 Lawful equivalence and its effect functors

**Definition 2.9** (Lawful Bell zigzag and groupoid completion). A lawful Bell zigzag from  $O$  to  $O'$  is a finite zigzag in  $\mathbf{Op}_{\mathcal{B}}$  whose legs are exactly recoverable simulations. Write  $O \simeq_{\mathcal{B}} O'$  when such a zigzag exists. Let  $\mathcal{G}_{\mathcal{B}}$  be the groupoid completion generated by these arrows, modulo identities, composition, and cancellation with formal inverses.

**Lemma 2.10** (Lawful Bell equivalence is an equivalence relation). *The relation  $\simeq_{\mathcal{B}}$  is reflexive, symmetric, and transitive, and  $\mathcal{G}_{\mathcal{B}}^{\text{can}}$  embeds fully into  $\mathcal{G}_{\mathcal{B}}$ .*

*Proof.* Use the empty zigzag, reversal, and concatenation. Canonical isomorphisms are generating exactly recoverable simulations.  $\square$

**Theorem 2.11** (Object-indexed governance-effect functors). *For each  $L \in \{D, A, T, C, \tau, U\}$ , the assignments*

$$O \mapsto \mathcal{X}_L(O), \quad f \mapsto F_f^L$$

*extend uniquely to a functor*

$$\mathcal{X}_L : \mathcal{G}_{\mathcal{B}} \longrightarrow \mathbf{Set}_{\cong}.$$

*For a zigzag  $h$ , the transport  $h_*^L$  is the corresponding composite of the  $F$  maps and their inverses and is independent of the chosen representative of  $h$ .*

*Proof.* The functor laws hold on  $\mathbf{Op}_{\mathcal{B}}$  by definition. The universal property of groupoid completion extends every functor that sends generating arrows to isomorphisms. Zigzag relations map to identity, composition, and inverse equations, so the induced transport is representative-independent.  $\square$

**Definition 2.12** (Diagonal effect transport). For an arrow  $h : O \rightarrow O'$  in  $\mathcal{G}_{\mathcal{B}}$  and a six-effect tuple  $e = (e_D, e_A, e_T, e_C, e_{\tau}, e_U)$  over  $O$ , define

$$h_*e := (h_*^De_D, h_*^Ae_A, h_*^Te_T, h_*^Ce_C, h_*^{\tau}e_{\tau}, h_*^Ue_U).$$

This is the diagonal action of the six functors of theorem 2.11.

**Theorem 2.13** (Lawful-localization closure). *Every arrow of  $\mathcal{G}_{\mathcal{B}}$  preserves the typed operational carrier, the admitted empirical table, and the reference target packet of its source and target. It also transports arbitrary candidate effect sections through the six functors of theorem 2.11; it does not identify a candidate section with the reference section. Consequently normalization, overlap consistency, and the reference target data are invariant under  $\simeq_{\mathcal{B}}$ , while candidate governance behavior is compared only after functorial transport. Global-section invariance is proved separately for the empirical lawful groupoid in theorem 4.4.*

*Proof.* Each generator preserves the typed carrier and satisfies the coordinatewise reference-section naturality equation of theorem 2.4; functorial transport carries arbitrary candidate sections under composition and formal inversion.  $\square$

## 2.3 Fixed domain and typed semantic presentations

**Definition 2.14** (Fixed Bell domain and target-practice packet). Fix the operational object  $O_P$  carrying the admitted witness and a reviewed token  $z_P \in \mathbf{Z}_P$ . Fix finite sets of mandatory report and reuse acts

$$\mathbf{MandRpt}_P \subseteq \mathbf{Rpt}_P, \quad \mathbf{MandReuse}_P \subseteq \mathbf{S}_P,$$

a mandatory continuation generator

$$\mathbf{MandCont}_P \subseteq \mathbf{S}_P \times \mathbf{S}_P, \quad \text{pr}_1(\mathbf{MandCont}_P) \cup \text{pr}_2(\mathbf{MandCont}_P) \subseteq \mathbf{MandReuse}_P,$$

a finite set of target-active standing-negative obligations

$$\text{MandNeg}_P \subseteq \mathcal{Z}_P,$$

and a channel-support map

$$\chi_P^{\text{ch}} : \text{MandRpt}_P \sqcup \text{MandReuse}_P \longrightarrow \mathcal{C}_P,$$

and a continuation-support relation

$$\text{CSupp}_P \subseteq (\text{MandRpt}_P \sqcup \text{MandReuse}_P) \times \text{MandCont}_P$$

with nonempty fiber over every mandatory report or reuse act. Fix a finite channel-status carrier  $\mathbf{H}_P$ , a pre-act channel-status map

$$\kappa_P^{\text{ch}} : \mathcal{C}_P \rightarrow \mathbf{H}_P,$$

and a set  $\text{DeclTr}_P \subseteq \{0, 1\}^{\mathbf{H}_P}$  of trace rules registered before the witness act. Require

$$\alpha_P = \bar{\alpha}_P \circ \kappa_P^{\text{ch}} \quad \text{for some } \bar{\alpha}_P \in \text{DeclTr}_P.$$

Let  $\pi_P^\tau : \text{Status} \rightarrow \text{Status}_P^{\text{gov}}$  be the target-bearing status quotient and let  $f_P \in \text{Status}_P^{\text{gov}}$  be its unique admitted final class. Define the permitted same-act status relation  $\text{SA}_P \subseteq \text{Status}^2$  by

$$(s, s') \in \text{SA}_P \iff \pi_P^\tau(s) = \pi_P^\tau(s') = f_P.$$

Define

$$s \approx_{\tau, P} s' \iff \pi_P^\tau(s) = \pi_P^\tau(s'), \quad \text{Final}_P(s) \iff \pi_P^\tau(s) = f_P.$$

The fixed target packet is

$$\begin{aligned} \Theta_P := & (\mathcal{C}_P, \alpha_P, \theta_P, z_P, \mathcal{Z}_P, \sigma_P, \eta_P, \mathcal{S}_P, \mathcal{K}_P, \text{Rpt}_P, \ell_P^R, \rho_P, \zeta_P, \\ & \mathbf{W}_P, \omega_P, \tau_P, \text{MandRpt}_P, \text{MandReuse}_P, \text{MandCont}_P, \text{MandNeg}_P, \\ & \ell_P^C, \chi_P^{\text{ch}}, \text{CSupp}_P, \mathbf{H}_P, \kappa_P^{\text{ch}}, \text{DeclTr}_P, \pi_P^\tau, f_P, \text{SA}_P). \end{aligned}$$

The fixed Bell domain and its lawful automorphism group are

$$\mathfrak{B}_P := ([O_P]_{\simeq_B}, \mathcal{M}_B, \Theta_P), \quad \text{Aut}_B(P) := \text{Aut}_{\mathcal{G}_B}(O_P).$$

For binary CHSH use,  $\text{CHSH} = E_{00} + E_{01} + E_{10} - E_{11}$  and Bell-local models satisfy  $|\text{CHSH}| \leq 2$  [2, 3, 4].

**Definition 2.15** (Target-active carriers and universal target relations). Let  $\text{S}_P^{\text{act}}$  be the least subset

of  $S_P$  containing  $\text{MandReuse}_P$  and every endpoint of a pair in  $\text{MandCont}_P$ . Set

$$\begin{aligned} \text{Rpt}_P^{\text{act}} &:= \text{MandRpt}_P, & Z_P^+ &:= \{z_P\} \cup \zeta_P(\text{Rpt}_P^{\text{act}} \sqcup S_P^{\text{act}}), & Z_P^{\text{act}} &:= Z_P^+ \cup \text{MandNeg}_P, \\ \mathcal{C}_P^{\text{act}} &:= \text{im}(\chi_P^{\text{ch}}), & W_P^{\text{act}} &:= W_P. \end{aligned}$$

Define the canonical active continuation relation by

$$\mathcal{K}_P^{\text{act}} := \text{RTC}_{S_P^{\text{act}}}(\text{MandCont}_P),$$

the least reflexive-transitive relation on  $S_P^{\text{act}}$  containing  $\text{MandCont}_P$ . The target is continuation-adequate iff

$$\mathcal{K}_P \cap (S_P^{\text{act}})^2 = \mathcal{K}_P^{\text{act}}.$$

For any relation  $R \subseteq (S_P^{\text{act}})^2$ , let  $\text{Reach}_R$  be its reflexive-transitive closure and define the target-bearing continuation-use function by

$$\text{Use}_P^{\text{cont}}(R; q) = 1 \iff \exists e \in \text{MandCont}_P [( \text{tok}_P(q), e) \in \text{CSupp}_P \wedge e \in \text{Reach}_R].$$

Define

$$R \equiv_{\text{cont}, P} R' \iff \text{Reach}_R = \text{Reach}_{R'} \quad \text{and} \quad \text{Use}_P^{\text{cont}}(R) = \text{Use}_P^{\text{cont}}(R').$$

Thus  $\equiv_{\text{cont}, P}$  is equality of target-bearing continuation work, not literal equality of governance-inert edge presentations. The active report/reuse carrier and query set are

$$U_P^{\text{act}} := \text{Rpt}_P^{\text{act}} \sqcup S_P^{\text{act}},$$

$$Q_{U, P}^{\text{act}} := \{q_{\text{report}}(r) : r \in \text{Rpt}_P^{\text{act}}\} \sqcup \{q_{\text{reuse}}(s) : s \in S_P^{\text{act}}\}.$$

Define the underlying mandatory-act map by

$$\text{tok}_P(q_{\text{report}}(r)) := \text{inl}(r), \quad \text{tok}_P(q_{\text{reuse}}(s)) := \text{inr}(s).$$

Define the unique active target-standing valuation

$$\sigma_P^{\text{act}}(z) := \begin{cases} 1, & z \in Z_P^+, \\ 0, & z \in \text{MandNeg}_P. \end{cases}$$

Target adequacy requires  $Z_P^+ \cap \text{MandNeg}_P = \emptyset$  and  $\sigma_P|_{Z_P^{\text{act}}} = \sigma_P^{\text{act}}$ . Coordinates outside these active carriers are governance-inert exactly when they induce no change in any target-bearing verdict, continuation, report, or reuse act; such coordinates are routed to skin by the imported null-effect theorem.

**Definition 2.16** (Target-active reduction of governance effects). For an effect section transported

to  $O_P$ , define

$$\begin{aligned}
\text{red}_{P,D}(d) &:= d, \\
\text{red}_{P,A}(a) &:= a|_{\mathcal{C}_P^{\text{act}}}, \\
\text{red}_{P,T}(t) &:= (t|_{q_{\text{cell}}}, t|_{Z_P^{\text{act}}}, t|_{W_P^{\text{act}}}), \\
\text{red}_{P,C}(c) &:= (c|_{q_{\text{real}}^{AB} \sqcup q_{\text{real}}^A \sqcup q_{\text{real}}^B}, [\{(s, s') \in (S_P^{\text{act}})^2 : c(q_{\text{cont}}(s, s')) = 1\}]_{\equiv_{\text{cont}, P}}), \\
\text{red}_{P,\tau}(v) &:= \pi_P^\tau(v(q_{\text{status}}(\eta_P))), \\
\text{red}_{P,U}(u) &:= u|_{Q_{U,P}^{\text{act}}}.
\end{aligned}$$

Write

$$\begin{aligned}
\text{red}_P(\text{Eff}_{\mathcal{B}}(G)) &:= (\text{red}_{P,D} E_D(G), \text{red}_{P,A} E_A(G), \text{red}_{P,T} E_T(G), \\
&\quad \text{red}_{P,C} E_C(G), \text{red}_{P,\tau} E_\tau(G), \text{red}_{P,U} E_U(G)).
\end{aligned}$$

These reductions retain exactly the target-bearing governance behavior. Equality outside them is not required for governance equivalence.

**Definition 2.17** (Same-domain preservation and scope change). A presentation preserves the Bell domain iff its operational object lies in  $[O_P]_{\simeq_{\mathcal{B}}}$  and the transported target packet equals  $\Theta_P$ . A modification changes scope iff its domain effect leaves that lawful orbit. A difference that changes no target-active reduced effect is governance-inert skin.

**Definition 2.18** (Candidate/reference base compatibility). For a candidate section  $b = (b_D, b_A, b_T, b_C, b_\tau, b_U)$  over the same typed carrier, define  $\text{BaseCompat}_O(b)$  to mean:

- (R1)  $b_D = d_O^{\text{ref}}$  on target, cover, channel-package, and act queries;
- (R2) all standing, continuation, report/reuse, and comparison coordinates are typed on the carriers and support maps of  $O$ ;
- (R3) the empirical-cell coordinates form a normalized overlap-compatible model on the cover of  $O$ ;
- (R4) report/reuse and realization coordinates obey the support maps and wing labels of  $O$ .

The reference section types the target against which a proposal is evaluated. The candidate section may differ from the reference governance verdicts; that difference is the object of the normal-form theorem, not a second silently competing packet.

**Definition 2.19** (Raw typed semantic presentation). A raw typed semantic presentation is

$$G = (|G|, g_0, O_G, m_G, \beta_G, \text{coh}_G),$$

where  $g_0 \in |G|$ ,  $O_G$  is an operational Bell object,  $m_G : |G| \rightarrow \mathbf{M}$  is presentation-level metric or coordinate metadata, and

$$\beta_G : |G| \longrightarrow \prod_{L \in \{D, A, T, C, \tau, U\}} \mathcal{X}_L(O_G)$$

is a typed answer section. Write  $\beta_G^L(g)$  for its  $L$ -coordinate. The predicate  $\text{coh}_G(g)$  is  $\text{BaseCompat}_{O_G}(\beta_G(g))$  together with

$$\begin{aligned}
\sum_{a,b} \beta_G^T(g)(q_{\text{cell}}(x, y, a, b)) &= 1 & \forall x, y, \\
\sum_b \beta_G^T(g)(q_{\text{cell}}(x, y, a, b)) &= \sum_b \beta_G^T(g)(q_{\text{cell}}(x, y', a, b)) & \forall x, a, y, y', \\
\sum_a \beta_G^T(g)(q_{\text{cell}}(x, y, a, b)) &= \sum_a \beta_G^T(g)(q_{\text{cell}}(x', y, a, b)) & \forall y, b, x, x', \\
\beta_G^C(g)(q_{\text{real}}^{AB}(x, y, a, b)) &= 1 \iff \beta_G^T(g)(q_{\text{cell}}(x, y, a, b)) > 0, \\
\beta_G^C(g)(q_{\text{real}}^A(x, a)) &= 1 \iff \sum_b \beta_G^T(g)(q_{\text{cell}}(x, y, a, b)) > 0, \\
\beta_G^C(g)(q_{\text{real}}^B(y, b)) &= 1 \iff \sum_a \beta_G^T(g)(q_{\text{cell}}(x, y, a, b)) > 0, \\
\beta_G^C(g)(q_{\text{cont}}(s, s')) &= 1 \implies \beta_G^U(g)(q_{\text{reuse}}(s)) = \beta_G^U(g)(q_{\text{reuse}}(s')) = 1, \\
\beta_G^U(g)(q) &= 1 \implies \beta_G^T(g)(q_{\text{stand}}(\bar{\zeta}_{O_G}(q))) = 1, \\
\beta_G^T(g)(q_{\text{cmp}}(w)) &= w(\beta_G^T(g)|_{q_{\text{cell}}}) & \forall w \in \mathbf{W}_{O_G}.
\end{aligned}$$

The six coordinate sections are additionally natural under the functors of theorem 2.11. The reference packet in  $O_G$  and the candidate packet  $\beta_G(g_0)$  are therefore related by the explicit two-level condition above rather than silently identified. The metadata  $m_G$  is not governance-bearing unless a later theorem licenses such a role. The internal carrier  $|G|$  may be an instrument algebra, causal model, process object, history structure, contextuality model, categorical presentation, or another mathematical package, provided it supplies this typed AASC-facing governance interface. Let  $\text{RawBellGov}_{\mathcal{B}}$  denote these records.

**Lemma 2.20** (Syntactic restriction to the six declared sorts). *Every  $G \in \text{RawBellGov}_{\mathcal{B}}$  has unique component maps*

$$\begin{aligned}
d_G : |G| &\rightarrow \mathcal{D}_B(O_G), & a_G : |G| &\rightarrow \mathcal{A}_B(O_G), & t_G : |G| &\rightarrow \mathcal{T}_B(O_G), \\
c_G : |G| &\rightarrow \mathcal{C}_B(O_G), & \tau_G : |G| &\rightarrow \mathcal{T}_B^{\text{act}}(O_G), & u_G : |G| &\rightarrow \mathcal{U}_B(O_G),
\end{aligned}$$

*obtained by projection from the displayed product. This lemma asserts only typed decomposition of the declared interface; it does not yet assert semantic exhaustiveness or exclude a seventh governance-bearing role.*

*Proof.* Products have unique coordinate projections. □

**Definition 2.21** (Typed Bell-governance interface). Let  $\text{BellGov}_{\mathcal{B}}$  be the class of raw typed semantic presentations equipped with the six projections of theorem 2.20. Define

$$E_D(G) := d_G(g_0), \quad E_A(G) := a_G(g_0), \quad E_T(G) := t_G(g_0),$$

$$E_C(G) := c_G(g_0), \quad E_\tau(G) := \tau_G(g_0), \quad E_U(G) := u_G(g_0),$$

and

$$\text{Eff}_\mathcal{B}(G) := (E_D(G), E_A(G), E_T(G), E_C(G), E_\tau(G), E_U(G)).$$

The functors  $\mathcal{X}_L$  define  $h \cdot G$  for every  $h : O_G \rightarrow O_{G'}$  in  $\mathcal{G}_\mathcal{B}$ .

**Definition 2.22** (Governance modification and derived act support). A governance modification  $\Phi : G \rightarrow G'$  is a carrier map taking  $g_0$  to  $g'_0$ , an arrow  $h : O_G \rightarrow O_{G'}$  in  $\mathcal{G}_\mathcal{B}$ , and six comparison relations between  $h_*^L(E_L(G))$  and  $E_L(G')$ . Define the same-act-support condition by

$$\text{SameActSupport}_\mathcal{B}(\Phi) \iff \ell^{\text{act}}(\eta_{O_G}) = \ell^{\text{act}}(\eta_{O_{G'}}).$$

The support value itself is the well-typed element

$$\text{ActSupp}_\mathcal{B}(\Phi) := \ell^{\text{act}}(\eta_{O_G}) \in \{A, B, AB\}.$$

Whenever  $\text{SameActSupport}_\mathcal{B}(\Phi)$  holds, this is also  $\ell^{\text{act}}(\eta_{O_{G'}})$ , so the value is common to both endpoints. Modifications compose by carrier-map composition, groupoid composition, and relational composition; support is derived from the fixed act-support map rather than supplied as an unconstrained label.

**Definition 2.23** (Target-active extensional governance equivalence). For same-target  $G_1, G_2 \in \text{BellGov}_\mathcal{B}$ , write  $G_1 \equiv_{\text{gov}} G_2$  iff there exist lawful arrows

$$h_i : O_{G_i} \longrightarrow O_P \quad (i = 1, 2)$$

and  $g \in \text{Aut}_\mathcal{B}(P)$  such that

$$g_* \text{red}_P(h_{1*} \text{Eff}_\mathcal{B}(G_1)) = \text{red}_P(h_{2*} \text{Eff}_\mathcal{B}(G_2)).$$

Equivalently, the two presentations induce the same diagonal  $\text{Aut}_\mathcal{B}(P)$ -orbit of target-bearing anchor, standing, realization/continuation, act-time, and report/reuse behavior. Internal carrier equality and inactive bookkeeping equality are neither necessary nor sufficient.

**Theorem 2.24** (Extensional governance equivalence). *The relation  $\equiv_{\text{gov}}$  is an equivalence relation on the presentations claiming the fixed target  $P$ .*

*Proof.* Transport every presentation into the  $O_P$  fiber and compare the diagonal  $\text{Aut}_\mathcal{B}(P)$ -orbits of the reduced tuples. Orbit equality is reflexive, symmetric, and transitive, and a different choice of transport changes only the chosen representative.  $\square$

**Definition 2.25** (Canonical Bell governance carrier). A canonical governance tuple over  $O$  is

$$C = (O, \delta_C, \alpha_C, \theta_C, \lambda_C, \tau_C, v_C, s_C),$$

where the six effect entries belong to  $\mathcal{D}_B(O)$ ,  $\mathcal{A}_B(O)$ ,  $\mathcal{T}_B(O)$ ,  $\mathcal{C}_B(O)$ ,  $\mathcal{T}_B^{\text{act}}(O)$ ,  $\mathcal{U}_B(O)$  and  $s_C$  is a skin component. Let  $\mathfrak{C}_B$  be the class of well-typed tuples and write  $E_D, E_A, E_T, E_C, E_\tau, E_U$  for the six projections. Set

$$\text{Eff}_B(C) := (E_D(C), E_A(C), E_T(C), E_C(C), E_\tau(C), E_U(C)).$$

**Definition 2.26** (Syntactic encoding). For  $G \in \text{BellGov}_B$ , define  $E_B(G) \in \mathfrak{C}_B$  by copying its six effect sections and placing all remaining internal carrier data in  $s_{E_B(G)}$ . Whenever  $G$  claims the fixed target  $P$ ,  $G \equiv_{\text{gov}} E_B(G)$  by the identity transport and equality of active reductions.

**Definition 2.27** (Orbit-valued active normalization). For a same-target  $C \in \mathfrak{C}_B$ , choose a lawful arrow  $h_C : O_C \rightarrow O_P$  and define

$$\text{OrbNorm}_B(C) := [\text{red}_P(h_{C*}\text{Eff}_B(C))]_{\text{Aut}_B(P)}.$$

No representative of the diagonal orbit is selected. Inactive coordinates and governance-inert internal carrier data are omitted from the normal form and routed to skin.

**Theorem 2.28** (Orbit-valued canonical normalization). *The orbit  $\text{OrbNorm}_B(C)$  is independent of the chosen transport. Moreover*

$$C_1 \equiv_{\text{gov}} C_2 \iff \text{OrbNorm}_B(C_1) = \text{OrbNorm}_B(C_2),$$

and lawful transport leaves  $\text{OrbNorm}_B$  invariant.

*Proof.* Two choices of transport differ by an element of  $\text{Aut}_B(P)$ . Equality of diagonal active orbits is exactly equality of target-bearing governance behavior in theorem 2.23.  $\square$

## 2.4 Physical occurrence and the neutral target

**Finite mechanized scaling.** In the Lean companion, joint probabilities are integer weights over denominator 32. The vector/Born witness has

$$S_{32} = 80, \quad S = \frac{80}{32} = \frac{5}{2},$$

while the Bell-local bound is  $|S_{32}| \leq 64$ . The exact positive-bound gap is  $16/32 = 1/2$ . This is a finite witness implementation, not a replacement for the general Bell domain.

**Definition 2.29** (Physical Bell correlation occurrence). A physical Bell correlation occurrence is the occurrence of a Bell-type operational family, including preparation, settings, detector events, outcomes, records, and estimated probabilities. Physical occurrence alone does not fix non-degenerate same-witness use.

**Definition 2.30** (Ontic non-degenerate Bell-witness determinacy). Let  $g_P^* := (d_P^*, \alpha_P^*, t_P^*, c_P^*, \tau_P^*, u_P^*)$  be the six reference effect sections induced by  $O_P$ . The predicate  $\text{NDBellWitness}(P)$  holds iff

$$\sigma_P|_{Z_P^{\text{act}}} = \sigma_P^{\text{act}},$$

the packet  $\Theta_P$  is fixed under  $\mathcal{G}_B$ , the active anchor verdict is  $\alpha_P^*|_{C_P^{\text{act}}}$ , the empirical-cell coordinates of  $t_P^*$  are  $P$ , and

$$\mathcal{K}_P \cap (\mathbf{S}_P^{\text{act}})^2 = \mathcal{K}_P^{\text{act}}, \quad \text{Final}_P(\tau_P^*),$$

while every act in  $\text{MandRpt}_P \sqcup \text{MandReuse}_P$  is licensed by  $u_P^*$  and supported by positive standing. Because every endpoint of  $\text{MandCont}_P$  lies in  $\text{MandReuse}_P$ , every state in the active continuation generator is among the licensed reuse states. These are the identity conditions of the already instantiated target practice. No locality, tensor, causal, or transport form is included in this definition.

**Proposition 2.31** (Physical occurrence is not yet Bell-witness determinacy). *For a Bell operational family  $P$ ,*

$$\text{PhysBellCorr}(P) \not\Rightarrow \text{NDBellWitness}(P).$$

*Proof.* Physical occurrence supplies the empirical events. It does not by itself supply the fixed lawful orbit, admitted anchor, positive standing token, continuation relation, act-time finality, or licensed report/reuse data required by theorem 2.30.  $\square$

## 2.5 Boundary traces for Bell witness channels

**Definition 2.32** (Bell boundary trace). A Bell boundary trace is a declared predicate

$$\text{Tr}_{Ch} : \mathcal{C}_P \rightarrow \{0, 1\}$$

on candidate setting/outcome interfaces, detector channels, calibration links, readout identifications, or coincidence rules. It is predeclared and non-generative in the formal sense  $\text{PreTr}_P(\text{Tr}_{Ch})$ , bivalent, invariant under  $\text{Aut}_B(P)$ , outcome-independent by its channel-status factorization, and wing-local in the typed sense that every automorphism preserves  $\ell_P^C$  and the trace is constant on the resulting fixed-wing channel orbits. It fixes anchor admission only; it does not create the Bell correlations or determine the comparison result.

**Theorem 2.33** (No arbitrary Bell singleton trace). *A singleton-valued channel predicate fixes a witness role only when its source is independently declared in the Bell-domain architecture and satisfies the trace conditions of theorem 2.32. Otherwise it is post-selection, hidden selector import, delayed repair, or scope change.*

*Proof.* Singleton cardinality alone is trivial to manufacture after the data are known. Prior declaration blocks after-the-fact selection; canonical stability fixes the same interface under lawful

re-expression; outcome independence blocks desired-result selection; wing locality blocks remote channel choice; and non-generativity blocks the predicate from creating its own standing.  $\square$

**Theorem 2.34** (Boundary-trace representation theorem). *Let  $I$  be a richer Bell-interface structure with an anchor-status map  $s_I : I \rightarrow \{0, 1\}$  satisfying prior declaration, bivalence, lawful transport invariance, outcome independence, wing locality, and non-generativity. Let  $I/\sim_I$  be the quotient by the lawful channel-isomorphism relation induced by  $\mathcal{G}_B$ . Then there is a unique trace*

$$\overline{\text{Tr}}_I : I/\sim_I \rightarrow \{0, 1\}$$

*such that  $s_I = \overline{\text{Tr}}_I \circ q_I$ . Every anchor-level standing verdict factors through this trace. The exclusion of a second anchor authority is supplied separately by same-scope closure.*

*Proof.* Lawful transport invariance makes  $s_I$  constant on  $\sim_I$ -classes, so the quotient universal property supplies the unique trace. The remaining trace conditions prevent result-conditioned or self-generating anchor status. Boundary-trace fixation and same-scope closure then exclude an independent second anchor authority [22, 17].  $\square$

**Theorem 2.35** (Bell boundary-trace admissibility). *Detector, setting, calibration, coincidence, instrument, or apparatus facts perform anchor-level Bell work only through the factorization of theorem 2.34. Further physical or mathematical structure may remain as support or standing content, but it is not a second anchor authority.*

*Proof.* The quotient trace is the complete bivalent anchor verdict of the richer interface. A further feature that leaves the trace unchanged performs no anchor work; a feature that changes anchor admission independently is a second classifier or a new domain. Exact same-scope closure excludes the former as governance-inert and the latter as independent same-domain authority.  $\square$

### 3 Kernel activation, semantic exhaustiveness, and primitive conservation

**Definition 3.1** (Vocabulary-neutral Bell-witness adequacy). For the fixed operational object  $O_P$ , define:

- (A1) target determinacy: every lawful transport of  $\Theta_P$  returns the same target packet;
- (A2) step evaluability:  $\sigma_P : Z_P \rightarrow \{0, 1\}$  assigns a bivalent verdict to every evaluated token;
- (A3) same-regime fidelity: a lawful transformation preserves all six typed effect sections, while a change in one of those sections is declared rather than hidden;

- (A4) act-time finality:  $\tau_P$  is fixed for the evaluated act  $\eta_P$ , and a later act may change status only under a fresh act identifier or a fresh target gate.

These are identity conditions of non-degenerate same-witness use, not optional AASC vocabulary.

**Definition 3.2** (Governed Bell-witness sameness). Define

$$\text{GovernedSameness}_{\mathcal{B}}(P) \iff A1(P) \wedge A2(P) \wedge A3(P) \wedge A4(P),$$

where A1–A4 are the four clauses of theorem 3.1. Thus governed sameness is exactly target determinacy, bivalent evaluability, same-regime fidelity, and act-time finality on the fixed witness.

**Definition 3.3** (Bell kernel roles). For a Bell witness  $P$ , write

$$K_{\mathcal{B}}(P) = \{\text{Adm}_P, \text{Stand}_P, \text{Ref}_P, \text{Irr}_P\}.$$

Reference is target determinacy, standing is bivalent eligibility, admissibility is same-regime fidelity, and irreversibility is act-time finality.

**Theorem 3.4** (Vocabulary-neutral adequacy forces the Bell kernel). *For every Bell witness family  $P$ ,*

$$\text{NDBellWitness}(P) \implies \text{GovernedSameness}_{\mathcal{B}}(P) \implies K_{\mathcal{B}}(P).$$

*Contrapositively, denial of a kernel role entails loss, refusal, or retyping of the non-degenerate same-witness target.*

*Proof.* The four clauses of theorem 3.1 perform exactly the four kernel roles. This is the Bell specialization of target adequacy forcing the kernel and non-degenerate construction forcing the full kernel [15, Lemma 2.6 and Theorem 2.34]. Removing one role either reinstates the same governance work under another name or removes the target condition it performed.  $\square$

**Definition 3.5** (Primitive Bell activation packet). The local activation packet consists only of:

- (H1) the one fixed target  $\Theta_P$  and witness act  $\eta_P$ ;
- (H2) the lawful orbit and object-indexed effect functors of theorem 2.11;
- (H3) the declared verdict set  $\{0, 1\}$  on the evaluated fragment;
- (H4) one declared boundary-admission map  $\alpha_P : \mathcal{C}_P \rightarrow \{0, 1\}$ ;
- (H5) act-time finality for  $\eta_P$ ;
- (H6) the six syntactically declared query sorts of theorem 2.3; and
- (H7) nontrivial comparison, report, continuation, and reuse data in  $\Theta_P$ .

No no-second-classifier conclusion and no semantic effect-exhaustion conclusion is included in this packet.

**Theorem 3.6** (Ontic Bell-witness determinacy activates the closure spine). *If  $\text{NDBellWitness}(P)$  holds, the primitive activation packet verifies the local hypotheses of the imported fixed-domain closure results. Consequently the Bell target inherits report preservation, no second same-domain admissibility invariant, semantic governance-locus exhaustion, anchor/tensor/skin effect-channel uniqueness, singleton admissible-interior closure, standing conservation, and no same-act repair.*

*Proof.* By theorem 3.4, the instantiated witness activates reference, standing, admissibility, and irreversibility. H1–H2 fix the same target and lawful carrier; H3 gives the evaluated bivalent interface; H4 supplies the one declared gate without presupposing that no rival gate exists; H5 supplies finality; H6 is a typed syntactic interface only; and H7 supplies nontrivial standing-bearing use. The exact upstream statements and Bell substitutions are recorded in section G. Their conclusions, rather than H4 or H6, exclude a second classifier and prove semantic exhaustiveness.  $\square$

**Theorem 3.7** (Kernel-activated semantic governance exhaustiveness). *Assume  $\text{NDBellWitness}(P)$ . Let  $G \in \text{BellGov}_B$  claim the same target. Every internal datum of  $|G|$  that changes Bell governance changes at least one of*

$$E_D(G), E_A(G), E_T(G), E_C(G), E_\tau(G), E_U(G).$$

*If it changes none, it is governance-inert skin. No seventh same-domain governance-bearing effect channel survives.*

*Proof.* The five-locus theorem classifies governance work by target/domain, anchor, standing/continuation, temporal status, or inert representation [17, Theorem 4.2]. ATS behavioral and element-effect exhaustion split the standing/continuation locus into standing-content and licensed continuation, separate report/reuse effects, and force each governance-bearing datum through one unique effect channel; null-effect data collapse to skin [18, Theorems 4A, 4C, 4D, 5, 5A, and 5C]. Same-scope operator closure excludes an independent selector or totalization channel [19]. These conclusions are available only after theorem 3.6.  $\square$

**Theorem 3.8** (Bell governance representation theorem). *Under  $\text{NDBellWitness}(P)$ , every same-target typed AASC-facing  $G \in \text{BellGov}_B$  is governance-equivalent to its syntactic encoding  $E_B(G) \in \mathfrak{C}_B$ , and every governance-bearing internal difference is recorded by exactly one of the six canonical effect coordinates.*

*Proof.* The identity arrow gives  $G \equiv_{\text{gov}} E_B(G)$  by theorem 2.26. The additional completeness statement follows from theorem 3.7: an internal difference that is not skin changes one and only one canonical effect channel.  $\square$

**Theorem 3.9** (Equivariance of semantic encoding). *For every  $h \in \mathcal{G}_{\mathcal{B}}$  and same-target  $G \in \text{BellGov}_{\mathcal{B}}$ ,*

$$E_B(h \cdot G) \equiv_{\text{gov}} h \cdot E_B(G).$$

*Proof.* Both sides contain the transported six effect sections. Functoriality is theorem 2.11; uniqueness of the governance-bearing effect channel is theorem 3.7.  $\square$

**Definition 3.10** (Semantic target claim, active exact realization, and target tuple). For  $G \in \text{BellGov}_{\mathcal{B}}$ , define  $\text{SemClaims}_{\mathcal{B}}(G, P)$  iff there exists  $h : O_G \rightarrow O_P$  in  $\mathcal{G}_{\mathcal{B}}$  such that the transported target identifier, measurement cover, channel carrier, active-carrier packet, and act identifier equal those of  $O_P$ , and the transported empirical-cell section equals  $P$  on the fixed cover and outcome carriers. The independent empirical lawful groupoid of theorem 4.3 is used for Bell-local/global-section invariance, not as a substitute for fixed-target Bell transport. Define

$$\text{SemRealizes}_{\mathcal{B}}(G, P) \iff \exists h : O_G \rightarrow O_P \text{ in } \mathcal{G}_{\mathcal{B}} \quad \text{red}_P(h_* \text{Eff}_{\mathcal{B}}(G)) = \text{red}_P(g_P^*).$$

For  $C \in \mathfrak{C}_{\mathcal{B}}$ ,  $\text{Claims}_{\mathcal{B}}(C, P)$  and  $\text{Realizes}_{\mathcal{B}}(C, P)$  are the corresponding conditions on its projections. Define

$$C_P^0 := (O_P, d_P^*, \alpha_P^*, t_P^*, c_P^*, \tau_P^*, u_P^*, s_0).$$

The target claim fixes which witness and active practice are under review. Exact realization means equality of target-bearing governance work; it does not require equality of inactive bookkeeping coordinates.

**Definition 3.11** (Standing, realization, and reuse projections). For a standing effect  $t \in \mathcal{T}_B(O)$  set

$$\text{cell}_t(x, y, a, b) := t(q_{\text{cell}}(x, y, a, b)), \quad \text{st}_t(z) := t(q_{\text{stand}}(z)), \quad \text{cmp}_t(w) := t(q_{\text{cmp}}(w)).$$

For  $c \in \mathcal{C}_B(O)$  set

$$\begin{aligned} \text{real}_c^{AB}(x, y, a, b) &:= c(q_{\text{real}}^{AB}(x, y, a, b)), \\ \text{real}_c^A(x, a) &:= c(q_{\text{real}}^A(x, a)), \quad \text{real}_c^B(y, b) := c(q_{\text{real}}^B(y, b)), \\ \text{cont}_c(s, s') &:= c(q_{\text{cont}}(s, s')). \end{aligned}$$

Define  $\text{rep}_u$  and  $\text{reuse}_u$  analogously on report/reuse queries. Define

$$\text{emp}(t) := \{(x, y, a, b) \mapsto \text{cell}_t(x, y, a, b)\},$$

and let  $\text{real}_c$  denote the triple of joint,  $A$ -local, and  $B$ -local support sets cut out by the corresponding realization coordinates of  $c$ .

**Definition 3.12** (Primitive target-practice adequacy predicates). For effects already transported to  $O_P$ , define the following predicates.

A trace  $a : \mathcal{C}_P \rightarrow \{0, 1\}$  is *predeclared and non-generative*, written  $\text{PreTr}_P(a)$ , iff

$$\exists \bar{a} \in \text{DeclTr}_P \quad a = \bar{a} \circ \kappa_P^{\text{ch}}.$$

Thus the trace factors only through the fixed pre-act channel-status carrier and cannot depend on empirical outcomes, standing, comparison, continuation, posterior support, or later reports. *Trace adequacy* holds iff  $\text{PreTr}_P(a)$ ,  $a$  is  $\text{Aut}_{\mathcal{B}}(P)$ -invariant, and

$$a|_{\mathcal{C}_P^{\text{act}}} = \alpha_P^*|_{\mathcal{C}_P^{\text{act}}}.$$

The displayed equality is reference to the same mandatory interface, not a tensor or locality conclusion.

*Standing candidacy* holds iff

$$\text{emp}(t) = e_P, \quad \text{cmp}_t(w) = w(e_P) \quad (w \in \mathbf{W}_P^{\text{act}}),$$

$$\text{st}_t(z_P) = 1, \quad \text{st}_t(z) = 0 \quad (z \in \text{MandNeg}_P).$$

Positive standing on the remaining active tokens is not separately assumed; it follows from mandatory supported use.

*Continuation candidacy* holds iff the joint and one-wing realization coordinates of  $c$  are the support and marginals of  $e_P$  and, for

$$R_c := \{(s, s') \in (\mathbf{S}_P^{\text{act}})^2 : \text{cont}_c(s, s') = 1\},$$

one has  $\text{MandCont}_P \subseteq R_c$ , with  $R_c$  reflexive and transitive. No upper inclusion in  $\mathcal{K}_P^{\text{act}}$  is assumed.

*Mandatory-use adequacy* holds iff

$$u(q_{\text{report}}(r)) = 1 \quad (r \in \text{MandRpt}_P), \quad u(q_{\text{reuse}}(s)) = 1 \quad (s \in \text{MandReuse}_P),$$

and every positive mandatory verdict has positive standing support through  $\bar{\zeta}_P$  and continuation support:

$$\text{Use}_P^{\text{cont}}(R_c; q) = 1 \quad (q \in \mathbf{Q}_{U,P}^{\text{act}}).$$

Write

$$\text{TraceAdeq}_P(a), \quad \text{StandCand}_P(t), \quad \text{ContCand}_P(c; t), \quad \text{UseAdeq}_P(u; t, c)$$

for these conditions, and define

$$\text{TimeAdeq}_P(v) \iff \text{Final}_P(v(q_{\text{status}}(\eta_P))).$$

The primitive conditions fix target identity, the admitted empirical witness, bivalent evaluated obligations, mandatory continuation generation, finality, and supported mandatory use. They do not assume tensor compatibility, Bell-locality status, causal form, transport form, or the absence of all mathematically definable incoming maps.

**Lemma 3.13** (Mandatory continuation closure). *If  $\text{ContCand}_P(c; t)$ , then*

$$\mathcal{K}_P^{\text{act}} \subseteq R_c.$$

*Proof.*  $R_c$  is reflexive and transitive and contains  $\text{MandCont}_P$ . By minimality of the reflexive-transitive closure, it contains  $\mathcal{K}_P^{\text{act}}$ .  $\square$

**Definition 3.14** (Primitive candidate-admissibility predicate). For  $C \in \mathfrak{C}_{\mathcal{B}}$ ,  $\text{Adm}_{\mathcal{B}}^{\text{cand}}(C; P)$  holds iff  $\text{Claims}_{\mathcal{B}}(C, P)$  and, after lawful transport to  $O_P$ :

(B1)  $\text{TraceAdeq}_P(E_A(C));$

(B2)  $\text{StandCand}_P(E_T(C));$

(B3)  $\text{ContCand}_P(E_C(C); E_T(C));$

(B4)  $\text{TimeAdeq}_P(E_{\tau}(C));$

(B5)  $\text{UseAdeq}_P(E_U(C); E_T(C), E_C(C));$

(B6) the domain effect remains in the fixed target orbit and the candidate introduces no fresh target identifier, act identifier, or anchor gate.

No clause quantifies over arbitrary incoming modifications. Standing creation, transfer, repair, and metric authority are excluded separately for kernel-licensed same-act transformations. This prevents an inadmissible source presentation from defeating occupancy merely because a mathematical map into the target can be written.

**Lemma 3.15** (Target-instantiation occupancy).

$$\text{NDBellWitness}(P) \implies \exists C \in \mathfrak{C}_{\mathcal{B}} (\text{Claims}_{\mathcal{B}}(C, P) \wedge \text{Adm}_{\mathcal{B}}^{\text{cand}}(C; P)).$$

*Proof.* The target's own six effect sections define  $C_P^0$ . The pre-act trace register gives  $\text{TraceAdeq}_P(\alpha_P^*)$ . Exact witness identity, the reviewed positive token, and the mandatory negative obligations give  $\text{StandCand}_P(t_P^*)$ . The target continuation relation is reflexive and transitive and contains  $\text{MandCont}_P$ , giving  $\text{ContCand}_P(c_P^*; t_P^*)$ . Finality and mandatory licensed use give the remaining clauses. The endpoint condition on  $\text{MandCont}_P$  ensures every active continuation state is a mandatory licensed reuse state. No no-flow theorem, tensor-compatibility theorem, or uniqueness conclusion is used to establish occupancy.  $\square$

**Theorem 3.16** (No cost-free Bell-target retyping). *A proposed replacement either preserves the target claim in theorem 3.10 or changes the empirical witness, target identifier, measurement cover, channel package, or act identifier and is a different target or domain. There is no third route on which a target-bearing change counts as preservation of the original witness.*

*Proof.* Target and empirical identity are the reference-bearing coordinates of the fixed practice. A change in any such coordinate is target or domain change. A proposal changing none of them remains a same-target proposal and is classified by the standing, continuation, temporal, and reuse theorems below; it is presentation skin only when none of its target-active governance effects changes.  $\square$

### 3.1 Bell standing-flow predicates

**Definition 3.17** (Kernel-licensed same-domain Bell act). A governance modification  $\Phi : G \rightarrow G'$  is a same-domain same-act modification when its lawful arrow transports source and target to the same target identifier, domain orbit, active channel carrier, and act identifier. Define

$$\text{FreshGate}(\Phi) \iff h_*^D E_D(G) \neq E_D(G') \vee h_*^A E_A(G) \neq E_A(G').$$

Define  $\text{SameActLic}_P(\Phi)$  to hold iff:

- (S1) both endpoint sections are coherent and their transported target, cover, channel-package, and act coordinates agree;
- (S2)  $\neg \text{FreshGate}(\Phi)$ ;
- (S3) the status pair belongs to  $\text{SA}_P$ ;
- (S4) the lawful carrier arrow preserves the declared act interface, wing maps, support maps, and pre-act trace register; and
- (S5) no fresh target token, gate token, channel-status rule, or act identifier occurs in the target section.

The modification is *kernel-licensed*, written  $\text{KLic}_{\mathcal{B}}(\Phi; P)$ , iff  $\text{SameActLic}_P(\Phi)$ , both endpoints claim  $P$ , and active standing is bivalent at both endpoints. This definition is independent of  $\text{Adm}_{\mathcal{B}}^{\text{cand}}$  and may be applied to a standing-negative source. It therefore states the exact domain on which standing conservation forbids same-act creation or repair without circularly assuming construction admissibility at both endpoints.

**Definition 3.18** (Standing preorder and derived act support). For standing effects on the same object, write  $t \preceq_P t'$  iff all empirical-cell and comparison coordinates agree and

$$\text{st}_t(z) = 1 \implies \text{st}_{t'}(z) = 1 \quad \forall z \in \mathcal{Z}_P^{\text{act}}.$$

Write  $t \prec_P t'$  when  $t \preceq_P t'$  and the active standing restrictions differ. For a modification  $\Phi : G \rightarrow G'$  with lawful transport  $h$ , define

$$\Delta_L(\Phi) := \{q \in \mathbf{Q}_L(O_P) : (h_*^L E_L(G))(q) \neq E_L(G')(q)\}.$$

The support of the evaluated act is the value  $\text{ActSupp}_{\mathcal{B}}(\Phi)$  from theorem 2.22. The modification is  $A$ -supported iff  $\text{SameActSupport}_{\mathcal{B}}(\Phi)$ ,  $\text{ActSupp}_{\mathcal{B}}(\Phi) = A$ ,  $\Delta_D(\Phi) = \Delta_A(\Phi) = \Delta_{\tau}(\Phi) = \emptyset$ , and empirical-cell and comparison coordinates remain fixed;  $B$ -support is defined symmetrically. Because  $\text{ActSupp}_{\mathcal{B}}$  is computed from the fixed act identifier and its declared support map, support is target data rather than an annotation chosen by the proposal being audited.

**Definition 3.19** (Standing-flow effects). For a same-domain modification  $\Phi : G \rightarrow G'$  with lawful transport  $h$ , define:

$\text{CreatesStand}_{\mathcal{B}}(\Phi)$

$\neg \text{FreshGate}(\Phi)$  and there is  $z$  with

$$\text{st}_{h_*^T E_T(G)}(z) = 0, \quad \text{st}_{E_T(G')}(z) = 1.$$

$\text{AmplifiesStand}_{\mathcal{B}}(\Phi)$

$\neg \text{FreshGate}(\Phi)$  and  $h_*^T E_T(G) \prec_P E_T(G')$ .

$\text{TransfersStand}_{\mathcal{B}}(\Phi)$

Either  $\Phi$  is  $A$ -supported and at least one of the following changes: standing on  $Z_P^B \cup Z_P^{AB}$ ; a  $B$ -local or joint realization coordinate  $q_{\text{real}}^B(y, b)$  or  $q_{\text{real}}^{AB}(x, y, a, b)$ ; a continuation coordinate  $q_{\text{cont}}(s, s')$  with  $s$  or  $s'$  in  $S_P^B \cup S_P^{AB}$ ; or report/reuse behavior on  $\text{Rpt}_P^B \cup \text{Rpt}_P^{AB} \cup S_P^B \cup S_P^{AB}$ . The  $B$ -supported case is symmetric with  $A$  and  $B$  exchanged. Joint  $AB$  coordinates count as distant governance whenever a one-wing act changes them.

$\text{RepairsSameAct}_{\mathcal{B}}(\Phi)$

The domain and act identifier are fixed, the act-time status changes, and there is  $z$  with

$$\text{st}_{h_*^T E_T(G)}(z) = 0, \quad \text{st}_{E_T(G')}(z) = 1.$$

$\text{MetricAuthorizes}_{\mathcal{B}}(\Phi)$

The transported target, active trace data, empirical cells, comparison profile, act identifier, wing maps, and support maps agree, while

$$m_G(g_0) \neq m_{G'}(g'_0)$$

and at least one target-bearing standing, continuation, report/reuse, or act-time verdict differs. Thus metric authority means that metric metadata is the only new discriminator of

a governance verdict; it is not defined by an anchor-bit change and is not excluded merely by  $\neg \text{FreshGate}(\Phi)$ .

These are explicit relations on transported, wing-indexed effect sections.

**Theorem 3.20** (Primitive same-domain standing-flow exclusion). *For every kernel-licensed same-domain Bell act  $\Phi$ ,*

$$\neg \text{CreatesStand}_{\mathcal{B}}(\Phi) \wedge \neg \text{AmplifiesStand}_{\mathcal{B}}(\Phi) \wedge \neg \text{TransfersStand}_{\mathcal{B}}(\Phi) \\ \wedge \neg \text{RepairsSameAct}_{\mathcal{B}}(\Phi) \wedge \neg \text{MetricAuthorizes}_{\mathcal{B}}(\Phi).$$

*Proof.* For a kernel-licensed same-act transformation, standing conservation excludes creation, amplification, transfer, and same-act restoration [15, Theorem 6.15]; the Bivalence Theorem gives the same-scope form [17, Theorem 3.6]. For the metric-authority branch, the declared nonmetric target packet and trace data are fixed while a target-bearing verdict changes. By theorem 3.7, that change must occupy one of the canonical governance-effect channels. Because metric metadata is neither part of the declared gate nor a reference-bearing target coordinate, governance-bearing dependence on it is an independent same-domain discriminator or second invariant; AMetric no-selector closure and no-second-invariant rigidity exclude that route [21, Theorem 2.5][17, Theorem 3.4]. If the metadata changes no target-bearing effect, it is skin. Same-scope operator closure independently excludes post-hoc repair, deferred admissibility, and any attempted metric selection operator [19, Corollaries 3.11–3.12 and Proposition 5.2]. The displayed predicates are exactly the Bell effect-section substitutions of those prohibited routes.  $\square$

**Theorem 3.21** (No additional target-bearing continuation work). *Assume  $\text{NDBellWitness}(P)$  and let  $C$  claim  $P$  and satisfy primitive candidate admissibility. If  $R_C$  is the active continuation relation of  $E_C(C)$ , then*

$$R_C \equiv_{\text{cont}, P} \mathcal{K}_P^{\text{act}}.$$

*Equivalently,  $R_C$  and  $\mathcal{K}_P^{\text{act}}$  generate the same target-bearing reachability and the same mandatory report/reuse support. An additional edge that changes neither consequence is presentation skin; an edge that changes one is a second continuation authority and is excluded.*

*Proof.* By theorem 3.13,  $\mathcal{K}_P^{\text{act}} \subseteq \text{Reach}_{R_C}$ . An additional edge that changes target-bearing reachability or  $\text{Use}_P^{\text{cont}}$  supplies continuation work not generated by the mandatory target transitions. Same-scope operator exhaustion and standing conservation exclude such independent authority [19, Theorem 4.5][17, Theorems 3.6 and 4.2]. If an additional edge changes neither reachability nor supported mandatory use, the null-effect theorem routes it to skin. Hence the two relations are  $\equiv_{\text{cont}, P}$ -equivalent.  $\square$

**Corollary 3.22** (No primitive distant completion). *A local act may alter a later physical or conditional state, but it cannot complete or confer standing on the distant wing as the same already evaluated Bell-witness act.*

*Proof.* Distant completion satisfies at least one of  $\text{CreatesStand}_{\mathcal{B}}$ ,  $\text{AmplifiesStand}_{\mathcal{B}}$ ,  $\text{TransfersStand}_{\mathcal{B}}$ , or  $\text{RepairsSameAct}_{\mathcal{B}}$ .  $\square$

**Import discipline.** The complete operative statements, PDF pages, primitive Bell hypotheses, substitutions, and local uses are reproduced in section G and in the project-level imported-theorem specialization ledger. No imported theorem generates the Bell probabilities, empirical model, or finite CHSH witness.

## 4 Bell empirical models, obstruction packet, and tensor compatibility

### 4.1 Empirical model and standing quotient

**Definition 4.1** (Bell measurement cover and empirical model). Let the measurement labels be  $\{A_x : x \in X\} \cup \{B_y : y \in Y\}$  and the contexts be

$$\mathcal{M}_{\mathcal{B}} = \{C_{xy} = \{A_x, B_y\} : x \in X, y \in Y\}.$$

The empirical model determined by  $P$  is

$$e_P = \{P_{xy}\}_{(x,y)}, \quad P_{xy}(a, b) = P(a, b \mid x, y).$$

It is compatible on overlaps iff context distributions induce the same one-wing marginal whenever contexts share a local measurement. In a Bell scenario this is operational no-signaling. Let  $\text{Emp}_{\mathcal{B}}$  denote the compatible empirical models on this fixed cover and outcome typing.

**Definition 4.2** (Exactly recoverable empirical simulation). Let  $e$  and  $e'$  be finite compatible empirical models, possibly on lawfully equivalent finite covers. An exactly recoverable empirical simulation consists of setting maps, contextwise outcome kernels and recovery kernels, together with mutually inverse affine maps

$$\hat{f} : \Delta(\Omega_e) \xrightarrow{\cong} \Delta(\Omega_{e'}), \quad \hat{r} : \Delta(\Omega_{e'}) \xrightarrow{\cong} \Delta(\Omega_e),$$

where  $\Omega_e = \prod_x A_x \times \prod_y B_y$ , such that

$$e' = f_* e, \quad e = r_* e',$$

and for every global distribution  $\nu$  and context  $(x, y)$ ,

$$\text{marg}_{f(x), f(y)}(\hat{f}\nu) = f_* \text{marg}_{x,y}(\nu),$$

with the analogous recovery equation. Thus global assignments and their context marginals are transported by the same empirical simulation.

**Definition 4.3** (Empirical lawful groupoid and equivalence). Let  $\mathcal{G}_{\mathcal{B}}^{\text{emp}}$  be the groupoid completion of the category of finite compatible empirical models and exactly recoverable empirical simulations. Write

$$e \simeq_{\mathcal{B}}^{\text{emp}} e'$$

iff there is an arrow from  $e$  to  $e'$  in  $\mathcal{G}_{\mathcal{B}}^{\text{emp}}$ . This relation is defined directly on empirical models and is independent of any governance packet.

**Theorem 4.4** (Empirical lawful equivalence and global-section invariance). *The relation  $\simeq_{\mathcal{B}}^{\text{emp}}$  is an equivalence relation. Every arrow  $g : e \rightarrow e'$  in  $\mathcal{G}_{\mathcal{B}}^{\text{emp}}$  induces a bijection*

$$\hat{g} : \Gamma(e) \xrightarrow{\cong} \Gamma(e').$$

*Consequently Bell-locality and Bell nonlocality are invariant under  $\simeq_{\mathcal{B}}^{\text{emp}}$ .*

*Proof.* Reflexivity, symmetry, and transitivity follow from the groupoid completion. If  $\nu \in \Gamma(e)$ , the commuting marginal equations give  $\text{marg}(\hat{g}\nu) = g_*e = e'$ , so  $\hat{g}\nu \in \Gamma(e')$ . The inverse arrow supplies the inverse map.  $\square$

**Definition 4.5** (Lawful Bell presentation class and transported active profile). Let  $\mathcal{P}_{\mathcal{B}}(P)$  be the class of presentations  $G \in \text{BellGov}_{\mathcal{B}}$  whose operational object lies in  $[O_P]_{\simeq_{\mathcal{B}}}$  and whose transported target identifier and act identifier are  $\theta_P$  and  $\eta_P$ . For  $G \in \mathcal{P}_{\mathcal{B}}(P)$  choose any lawful arrow  $h_G : O_G \rightarrow O_P$  and define

$$\text{Prof}_{\mathcal{B}}^P(G) := \left[ \text{red}_P(h_{G*}\text{Eff}_{\mathcal{B}}(G)) \right]_{\text{Aut}_{\mathcal{B}}(P)}.$$

If  $h'_G$  is another choice, then  $h'_G h_G^{-1} \in \text{Aut}_{\mathcal{B}}(P)$ , so the orbit is choice-independent. The profile contains exactly the active anchor, empirical/comparison/standing, realization/continuation, final-status, and mandatory report/reuse behavior. Inactive coordinates are absent and cannot obstruct collapse of governance-equivalent presentations.

**Theorem 4.6** (Bell-profile completeness). *Assume  $\text{NDBellWitness}(P)$ . Every same-domain target-bearing Bell verdict on a presentation in  $\mathcal{P}_{\mathcal{B}}(P)$  is determined by  $\text{Prof}_{\mathcal{B}}^P$ . If two lawful presentations have equal transported active profiles, every remaining difference is governance-inert skin; if a difference changes the target practice, it changes at least one active reduced coordinate or leaves the fixed domain.*

*Proof.* Kernel-activated semantic exhaustiveness is theorem 3.7. The diagonal profile contains the active reductions of all six effect channels in one common fiber. The imported null-effect theorem routes differences outside the active target-bearing fragment to skin.  $\square$

**Definition 4.7** (Bell standing equivalence). For  $G, G' \in \mathcal{P}_{\mathcal{B}}(P)$ , write

$$G \sim_{\mathcal{B}} G' \iff \text{Prof}_{\mathcal{B}}^P(G) = \text{Prof}_{\mathcal{B}}^P(G').$$

The class  $[G]_{\sim_{\mathcal{B}}}$  is the Bell standing class.

**Corollary 4.8** (Exact Bell standing quotient). *Any presentation carrying exactly and only the transported six-effect Bell profile is uniquely isomorphic to*

$$\mathcal{P}_{\mathcal{B}}(P)/\sim_{\mathcal{B}}.$$

*The substantive result is profile completeness in theorem 4.6; the present statement is the quotient universal-property corollary.*

## 4.2 Bell-local completion and global-section obstruction

**Definition 4.9** (Global-section set). For a finite empirical model  $e_P$ , define

$$\Gamma(e_P) = \left\{ \nu \in \Delta \left( \prod_{x \in X} A_x \times \prod_{y \in Y} B_y \right) : \text{marg}_{x,y}(\nu) = P_{xy} \text{ for every } (x, y) \right\}.$$

**Definition 4.10** (Marginal fiber, Bell-local subfiber, and comparison profile). Let

$$\pi_{\mathcal{B}}(e) = (\{e|_{A_x}\}_{x \in X}, \{e|_{B_y}\}_{y \in Y}),$$

and define

$$\mathcal{F}_{\mathcal{B}}(P) = \{e \in \text{Emp}_{\mathcal{B}} : \pi_{\mathcal{B}}(e) = \pi_{\mathcal{B}}(e_P)\}, \quad \mathcal{L}_{\mathcal{B}}(P) = \{e \in \mathcal{F}_{\mathcal{B}}(P) : \Gamma(e) \neq \emptyset\}.$$

Let  $\mathcal{W}_{\mathcal{B}}(P)$  be the declared Bell-comparison profile—the CHSH functional and relabeling-complete family in the binary case.

**Definition 4.11** (Independently sufficient local standing). A Bell witness has independently sufficient local standing iff there exist a probability space  $(\Lambda, \Sigma, \mu)$ , with  $\mu$  independent of the settings, and normalized measurable stochastic kernels

$$P_A(a \mid x, \lambda), \quad P_B(b \mid y, \lambda),$$

such that local responses exhaust the standing-bearing contribution and

$$P(a, b \mid x, y) = \int_{\Lambda} P_A(a \mid x, \lambda) P_B(b \mid y, \lambda) d\mu(\lambda).$$

This is the AASC structural counterpart of Bell-local factorization.

**Theorem 4.12** (Local standing, Bell-locality, and global completion). *For a finite Bell scenario, independently sufficient local standing yields Bell-local factorization and a global section. Conversely, every global section is a convex mixture of deterministic global assignments and supplies a Bell-local model. In the binary CHSH scenario, these conditions are equivalent to the complete CHSH inequality family [5, 8].*

*Proof.* For each  $\lambda$ , define

$$\nu_\lambda(\mathbf{a}, \mathbf{b}) = \prod_x P_A(a_x | x, \lambda) \prod_y P_B(b_y | y, \lambda),$$

and integrate over  $\mu$ . Marginalizing to  $(x, y)$  recovers the Bell-local factorization. Conversely, use a global assignment as the hidden variable, with deterministic local responses given by its coordinates. Fine's theorem supplies the binary CHSH equivalence.  $\square$

**Corollary 4.13** (Bell nonlocality is global-section obstruction). *If the admitted witness violates the relevant Bell inequality, then  $\Gamma(e_P) = \emptyset$  and independently sufficient local standing is unavailable.*

### 4.3 Exact context realization before compatibility typing

**Definition 4.14** (Witness context realization family). For every context  $(x, y)$  define

$$\mu_{xy}^P(a, b) := P(a, b | x, y), \quad \text{Real}_{xy}(P) := \text{supp } \mu_{xy}^P.$$

Define the local and aggregate realization carriers

$$\text{Real}_x^A(P) := \text{supp} \left( \sum_b P(a, b | x, y) \right), \quad \text{Real}_y^B(P) := \text{supp} \left( \sum_a P(a, b | x, y) \right),$$

where overlap consistency makes these independent of the remote setting, and

$$\text{Real}^A(P) := \bigsqcup_x \{x\} \times \text{Real}_x^A(P), \quad \text{Real}^B(P) := \bigsqcup_y \{y\} \times \text{Real}_y^B(P),$$

$$\text{Real}(P) := \bigsqcup_{(x,y) \in X \times Y} \{(x, y)\} \times \text{Real}_{xy}(P).$$

Let  $r_{xy}^{A,P}$  and  $r_{xy}^{B,P}$  be the ordinary marginalization maps. The target continuation relation is the already declared relation

$$\mathcal{K}_B^P := \mathcal{K}_P \subseteq \mathbf{S}_P \times \mathbf{S}_P.$$

These objects are defined directly from  $O_P$  before  $T_B(P)$  is introduced.

**Theorem 4.15** (Context realization certificate). *The family  $\{\mu_{xy}^P\}$  is normalized, has the admitted one-wing marginals, recovers the complete witness, is compatible on overlaps, and transforms*

covariantly under the functor  $\mathcal{K}_T$ . Any exact realization family for the same Bell profile agrees with it after lawful transport.

*Proof.* Existence and exact recovery are definitional. Normalization is normalization of  $P_{xy}$ ; overlap compatibility is no-signaling. Covariance is the empirical-cell component of theorem 2.11. Exactness fixes every context measure, leaving only lawful transport.  $\square$

#### 4.4 Bell joint obstruction packet

**Definition 4.16** (Bell obstruction diagram and packet). Define the pointed Bell obstruction diagram

$$\mathbf{J}(P) := (e_P, \pi_{\mathcal{B}}(e_P), \mathcal{F}_{\mathcal{B}}(P), \mathcal{L}_{\mathcal{B}}(P), \Gamma(e_P)),$$

with the evident marginal, inclusion, and global-section maps. Every arrow  $g : e \rightarrow e'$  in  $\mathcal{G}_{\mathcal{B}}^{\text{emp}}$  induces affine bijections

$$g_*^{\text{fb}} : \mathcal{F}_{\mathcal{B}}(e) \xrightarrow{\cong} \mathcal{F}_{\mathcal{B}}(e'), \quad g_*^{\text{loc}} : \mathcal{L}_{\mathcal{B}}(e) \xrightarrow{\cong} \mathcal{L}_{\mathcal{B}}(e'),$$

together with the global-section bijection of theorem 4.4. The comparison profile is added only at the governance and compatibility layer, because it is not a field of the empirical simulation category. Let

$$\text{Job}_{\mathcal{B}}(P) := [\mathbf{J}(P)]_{\mathcal{G}_{\mathcal{B}}^{\text{emp}}}$$

be the isomorphism class of this pointed diagram. For Bell-nonlocal  $P$ ,

$$e_P \in \mathcal{F}_{\mathcal{B}}(P) \setminus \mathcal{L}_{\mathcal{B}}(P), \quad \Gamma(e_P) = \emptyset.$$

The packet is functorial up to empirical lawful isomorphism. Its asserted irreducibility is governance-role irreducibility, not coordinate minimality.

**Definition 4.17** (Reduced Bell obstruction equivalence). On  $\mathcal{F}_{\mathcal{B}}(P)$  let  $R_{\text{loc}}$  be the relation

$$e R_{\text{loc}} e' \iff (e \simeq_{\mathcal{B}}^{\text{emp}} e') \vee (e, e' \in \mathcal{L}_{\mathcal{B}}(P)).$$

Let  $\approx_{\text{loc}}$  be the smallest equivalence relation containing  $R_{\text{loc}}$ . Equivalently, it is the equivalence closure of lawful Bell equivalence together with the complete relation on the Bell-local subfiber. Define

$$\text{Ob}_{\mathcal{B}}(P) := [e_P]_{\approx_{\text{loc}}} \in \mathcal{F}_{\mathcal{B}}(P) / \approx_{\text{loc}}.$$

The common class of  $\mathcal{L}_{\mathcal{B}}(P)$  is the distinguished base point.

**Lemma 4.18** (Local base point is separated from Bell-nonlocal classes). *Empirical lawful equivalence preserves existence of a global section. Hence no Bell-nonlocal empirical model is  $\approx_{\text{loc}}$ -equivalent to the Bell-local base point.*

*Proof.* Global-section existence is invariant under  $\simeq_{\mathcal{B}}^{\text{emp}}$  by theorem 4.4. Every generating pair added beyond lawful equivalence lies entirely inside  $\mathcal{L}_{\mathcal{B}}(P)$ . A finite chain beginning in the Bell-local subfiber therefore cannot leave it.  $\square$

**Theorem 4.19** (Functorial canonicity and role nontriviality). *The assignment  $e \mapsto \mathbf{J}(e)$  is functorial on  $\mathcal{G}_{\mathcal{B}}^{\text{emp}}$  and sends every arrow to an isomorphism of pointed obstruction diagrams. Hence  $\text{JOb}_{\mathcal{B}}(P)$  is invariant as an isomorphism class. The reduced class  $\text{Ob}_{\mathcal{B}}(P)$  is invariant under the induced fiber isomorphisms. If  $P$  is Bell-nonlocal,  $\text{Ob}_{\mathcal{B}}(P)$  is not the base point, and no independently sufficient local-standing model has the same packet or reduced obstruction class.*

*Proof.* The affine global-assignment bijection commutes with all context marginals, hence transports the marginal fiber. Global-section invariance transports the Bell-local subfiber and the distinguished base point. Identity and composition follow from the empirical functor laws. Bell nonlocality gives  $e_P \notin \mathcal{L}_{\mathcal{B}}(P)$ , so theorem 4.18 makes its class non-base-point. Independently sufficient local standing supplies a global section by theorem 4.12 and therefore lies at the base point.  $\square$

## 4.5 Raw compatibility data, certification, and exact equivalence

**Definition 4.20** (Raw Bell compatibility data). A raw Bell compatibility datum for the fixed witness  $P$  is

$$T^{\text{raw}} = (\nu^T, \sigma_T, r^{A,T}, r^{B,T}, \mathcal{R}_T^A, \mathcal{R}_T^B, \mathcal{R}_T^{AB}, \mathcal{K}_T, \text{cov}_T, \omega_T, \rho_T) \in \text{BellCompat}_{\mathcal{B}}^{\text{raw}}(P),$$

where:

- (R1)  $\nu_{xy}^T : A_x \times B_y \rightarrow [0, 1]$  is a context distribution;
- (R2)  $\sigma_T : Z_P \rightarrow \{0, 1\}$  is the standing-token valuation;
- (R3)  $r_{xy}^{A,T}, r_{xy}^{B,T}$  are the marginal maps;
- (R4)  $\mathcal{R}_T^A \subseteq \bigsqcup_x \{x\} \times A_x$ ,  $\mathcal{R}_T^B \subseteq \bigsqcup_y \{y\} \times B_y$ , and  $\mathcal{R}_T^{AB} \subseteq \bigsqcup_{x,y} \{(x, y)\} \times A_x \times B_y$  are the wing-local and joint realization supports;
- (R5)  $\mathcal{K}_T \subseteq S_P \times S_P$  is the continuation relation;
- (R6)  $\text{cov}_T(g)$  is a coherent transport isomorphism for every  $g \in \text{Aut}_{\mathcal{B}}(P)$ ;
- (R7)  $\omega_T : W_P \rightarrow \{-1, 0, 1\}$  is the Bell-comparison profile; and
- (R8)  $\rho_T : \text{Rpt}_P \sqcup S_P \rightarrow \{0, 1\}$  is the report/reuse verdict.

The domain orbit, anchor effect, and act-time status are fixed externally by  $P$  and are not duplicated inside the compatibility datum.

**Definition 4.21** (Compatibility certificate predicate). For  $T^{\text{raw}} \in \text{BellCompat}_{\mathcal{B}}^{\text{raw}}(P)$ , define

$$\text{CompatCert}_P(T^{\text{raw}})$$

to hold iff:

- (C1) every  $\nu_{xy}^T$  is normalized;
- (C2) the marginal maps agree on overlaps;
- (C3)  $\sigma_T = \sigma_P$ ;
- (C4)  $\mathcal{R}_T^{AB} = \bigsqcup_{x,y} \{(x, y)\} \times \text{supp } \nu_{xy}^T$  and  $\mathcal{R}_T^A, \mathcal{R}_T^B$  are its one-wing marginal supports;
- (C5)  $\mathcal{K}_T$  is a relation on  $\mathcal{S}_P^{\text{act}}$  with  $\mathcal{K}_T \equiv_{\text{cont}, P} \mathcal{K}_P^{\text{act}}$ ; target and act identity are fixed by the ambient  $O_P$  fiber rather than reasserted as an untyped property of the relation;
- (C6)  $\text{cov}_T$  satisfies the identity and composition laws for  $\text{Aut}_{\mathcal{B}}(P)$  under  $\mathcal{X}_T$  and  $\mathcal{X}_C$ ;
- (C7)  $\omega_T$  is the comparison profile computed from  $\nu^T$  on the declared test family; and
- (C8) for every  $q \in \mathcal{Q}_{U,P}^{\text{act}}$ , a positive report or reuse verdict satisfies  $\sigma_T(\bar{\zeta}_P(q)) = 1$  and  $\text{Use}_P^{\text{cont}}(\mathcal{K}_T; q) = 1$ . No certificate condition is imposed on inactive report/reuse coordinates; they are skin unless they alter an active consequence.

**Definition 4.22** (Certified Bell compatibility structure). A certified Bell compatibility structure is

$$T = (T^{\text{raw}}, \kappa_T) \in \text{BellCompat}_{\mathcal{B}}(P), \quad \kappa_T : \text{CompatCert}_P(T^{\text{raw}}).$$

Proof terms are not governance-bearing data; only the extensional fields they certify enter equivalence.

**Proposition 4.23** (Type-level compatibility role separation). *A raw compatibility datum has no domain-effect, anchor-effect, metric-authority, or act-time field. Hence any such authority attributed to  $T$  is not encoded by the compatibility type and must enter through an external governance coordinate. Under kernel-activated effect-channel uniqueness, it is either the fixed target value, skin, a second classifier, repair, or scope change.*

*Proof.* Inspect the fields of theorem 4.20. The semantic classification follows from theorem 3.7.  $\square$

**Definition 4.24** (Separated compatibility effects). For  $T \in \text{BellCompat}_{\mathcal{B}}(P)$ , define three effect

sections:

$$\begin{aligned}
\text{stEff}_P(T)(q_{\text{cell}}(x, y, a, b)) &:= \nu_{xy}^T(a, b), \\
\text{stEff}_P(T)(q_{\text{stand}}(z)) &:= \sigma_T(z), \\
\text{stEff}_P(T)(q_{\text{cmp}}(w)) &:= \omega_T(w), \\
\text{ctEff}_P(T)(q_{\text{real}}^{AB}(x, y, a, b)) &:= \mathbf{1}_{\mathcal{R}_T^{AB}}(x, y, a, b), \\
\text{ctEff}_P(T)(q_{\text{real}}^A(x, a)) &:= \mathbf{1}_{\mathcal{R}_T^A}(x, a), \\
\text{ctEff}_P(T)(q_{\text{real}}^B(y, b)) &:= \mathbf{1}_{\mathcal{R}_T^B}(y, b), \\
\text{ctEff}_P(T)(q_{\text{cont}}(s, s')) &:= \mathbf{1}_{\mathcal{K}_T}(s, s'), \\
\text{useEff}_P(T)(q_{\text{report}}(r)) &:= \rho_T(\text{inl}(r)), \\
\text{useEff}_P(T)(q_{\text{reuse}}(s)) &:= \rho_T(\text{inr}(s)).
\end{aligned}$$

Set

$$\text{CompProf}_P(T) := (\text{stEff}_P(T), \text{ctEff}_P(T), \text{useEff}_P(T)).$$

Anchor, domain, and act-time information is deliberately absent from  $\text{stEff}_P$  and supplied separately by the fixed target.

**Definition 4.25** (Exact faithful compatibility realization). For  $T \in \text{BellCompat}_{\mathcal{B}}(P)$ , define  $\text{ExactFaithful}_P(T)$  to hold iff the following target-bearing conditions hold after lawful transport:

- (E1)  $\nu_{xy}^T = P_{xy}$  for every context;
- (E2) the marginal maps are  $r_{xy}^{A,P}, r_{xy}^{B,P}$ ;
- (E3)  $\sigma_T|_{\mathcal{Z}_P^{\text{act}}} = \sigma_P^{\text{act}}$ ;
- (E4)  $\mathcal{R}_T^{AB} = \text{Real}(P)$ ,  $\mathcal{R}_T^A = \text{Real}^A(P)$ , and  $\mathcal{R}_T^B = \text{Real}^B(P)$ ;
- (E5)  $\mathcal{K}_T \equiv_{\text{cont}, P} \mathcal{K}_P^{\text{act}}$ ;
- (E6)  $\text{cov}_T$  is the target  $\text{Aut}_{\mathcal{B}}(P)$ -covariance family on the active profile;
- (E7)  $\omega_T|_{\mathcal{W}_P^{\text{act}}} = \omega_P|_{\mathcal{W}_P^{\text{act}}}$ ;
- (E8)  $\text{useEff}_P(T)$  agrees with the target report/reuse effect on  $\mathcal{Q}_{U,P}^{\text{act}}$ ; and
- (E9)  $\Gamma(e_P) = \emptyset$ .

Inactive differences are not compatibility differences unless they alter an active target-bearing consequence.

**Definition 4.26** (Target-active compatibility equivalence). For  $T_1, T_2 \in \text{BellCompat}_{\mathcal{B}}(P)$ , write  $T_1 \equiv_{\text{comp}} T_2$  iff there is  $g \in \text{Aut}_{\mathcal{B}}(P)$  such that

$$(\text{red}_{P,T}, \text{red}_{P,C}, \text{red}_{P,U})(g_* \text{CompProf}_P(T_1)) = (\text{red}_{P,T}, \text{red}_{P,C}, \text{red}_{P,U})(\text{CompProf}_P(T_2)).$$

Proof fields and inactive coordinates are identified whenever they certify the same target-bearing transported behavior.

**Lemma 4.27** (Compatibility equivalence is an equivalence relation). *The relation  $\equiv_{\text{comp}}$  is reflexive, symmetric, and transitive.*

*Proof.* Use the identity, inverse, and composite automorphisms in  $\text{Aut}_{\mathcal{B}}(P)$  and the effect-functor laws.  $\square$

**Proposition 4.28** (Compatibility profile detects compatibility equivalence). *For  $T_1, T_2 \in \text{BellCompat}_{\mathcal{B}}(P)$ ,  $T_1 \equiv_{\text{comp}} T_2$  iff their transported active standing, continuation, and report/reuse reductions agree.*

*Proof.* This is the extensional target-active definition of  $\equiv_{\text{comp}}$ .  $\square$

**Definition 4.29** (Compatibility-to-governance embedding). Define

$$\iota_P(T) := (O_P, d_P^*, \alpha_P^*, \text{stEff}_P(T), \text{ctEff}_P(T), \tau_P^*, \text{useEff}_P(T), s_0) \in \mathfrak{C}_{\mathcal{B}}.$$

Then

$$T_1 \equiv_{\text{comp}} T_2 \iff \iota_P(T_1) \equiv_{\text{gov}} \iota_P(T_2).$$

**Definition 4.30** (Exact compatibility class and exact standing effects). Let

$$\text{BellCompat}_{\mathcal{B}}(P)^{\text{ex}}(P) := \{T \in \text{BellCompat}_{\mathcal{B}}(P) : \text{ExactFaithful}_P(T)\}$$

and

$$\mathcal{T}_B^{\text{exact}}(P) := \{\text{red}_{P,T}(\text{stEff}_P(T)) : T \in \text{BellCompat}_{\mathcal{B}}(P)^{\text{ex}}(P)\}.$$

For  $t \in \mathcal{T}_B^{\text{exact}}(P)$  define

$$\bar{\chi}_P(t) := [T]_{\equiv_{\text{comp}}}$$

for any exact faithful  $T$  with  $\text{red}_{P,T}(\text{stEff}_P(T)) = t$ .

**Lemma 4.31** (Exact standing effect determines the compatibility class). *The map  $\bar{\chi}_P$  is well-defined. If  $T_1, T_2$  are exact faithful and their active standing reductions agree after lawful transport, then  $T_1 \equiv_{\text{comp}} T_2$ .*

*Proof.* Exact faithfulness fixes the active continuation and report/reuse reductions. Equality of the active standing reductions therefore gives equality of the full target-active compatibility profiles.  $\square$

**Definition 4.32** (Canonical raw Bell compatibility data). Define  $T_B^{\text{raw}}(P) \in \text{BellCompat}_B^{\text{raw}}(P)$  by

$$\begin{aligned} \nu_{xy}^{T_B^{\text{raw}}(P)} &:= \mu_{xy}^P, & \sigma_{T_B^{\text{raw}}(P)} &:= \sigma_P, \\ r_{xy}^{A, T_B^{\text{raw}}(P)} &:= r_{xy}^{A, P}, & r_{xy}^{B, T_B^{\text{raw}}(P)} &:= r_{xy}^{B, P}, \\ \mathcal{R}_{T_B^{\text{raw}}(P)}^{AB} &:= \text{Real}(P), & \mathcal{K}_{T_B^{\text{raw}}(P)} &:= \mathcal{K}_P^{\text{act}}, \\ \mathcal{R}_{T_B^{\text{raw}}(P)}^A &:= \text{Real}^A(P), & \mathcal{R}_{T_B^{\text{raw}}(P)}^B &:= \text{Real}^B(P), \\ \omega_{T_B^{\text{raw}}(P)} &:= \omega_P, & \rho_{T_B^{\text{raw}}(P)} &:= \rho_P. \end{aligned}$$

Its covariance field is the restriction of functorial transport to  $\text{Aut}_B(P)$ .

**Theorem 4.33** (Compatibility certification). *For a compatible Bell-nonlocal empirical model  $P$ ,*

$$\text{CompatCert}_P(T_B^{\text{raw}}(P)).$$

*Proof.* C1–C4 are theorem 4.15 together with the fixed valuation  $\sigma_P$ . For C5, the canonical datum uses  $\mathcal{K}_P^{\text{act}}$  itself, so it is a relation on the active carrier and is reflexive-transitive by construction; reflexivity of  $\equiv_{\text{cont}, P}$  gives  $\mathcal{K}_{T_B^{\text{raw}}(P)} \equiv_{\text{cont}, P} \mathcal{K}_P^{\text{act}}$ . C6 is the  $\text{Aut}_B(P)$  restriction of theorem 2.11; C7 is the declared Bell comparison profile; and C8 is the target-active report/reuse discipline supplied by  $\text{NDBellWitness}(P)$ .  $\square$

**Definition 4.34** (Canonical certified Bell compatibility object). Define

$$T_B(P) := (T_B^{\text{raw}}(P), \kappa_P) \in \text{BellCompat}_B(P),$$

where  $\kappa_P$  is the proof of theorem 4.33.

**Lemma 4.35** (Canonical object is exact faithful).

$$\text{ExactFaithful}_P(T_B(P)).$$

*Proof.* E1–E8 are the defining fields of  $T_B^{\text{raw}}(P)$  and E9 is Bell nonlocality from theorem 4.13.  $\square$

**Theorem 4.36** (Exact compatibility uniqueness). *If  $T \in \text{BellCompat}_B(P)$  and  $\text{ExactFaithful}_P(T)$ , then*

$$T \equiv_{\text{comp}} T_B(P), \quad \iota_P(T) \equiv_{\text{gov}} \iota_P(T_B(P)).$$

*Proof.* Exactness fixes the full compatibility profile after lawful transport. The first conclusion is extensionality; the second is theorem 4.29. The substantive result is certification of the canonical object.  $\square$

**Definition 4.37** (Semantic target deviations and terminal discharge predicates). For  $G \in \text{BellGov}_B$ , let  $\text{emp}(E_T(G))$  be the empirical model given by the cell coordinates of  $E_T(G)$ . Define:

AnchorDeviation( $G$ )

$\text{SemClaims}_{\mathcal{B}}(G, P)$  and  $\text{red}_{P,A}(E_A(G)) \neq \alpha_P^*|_{\mathcal{C}_P^{\text{act}}}$  after lawful transport.

ContinuationDeviation( $G$ )

$\text{SemClaims}_{\mathcal{B}}(G, P)$  and  $\text{red}_{P,C}(E_C(G)) \neq \text{red}_{P,C}(\text{ctEff}_P(T_B(P)))$ .

ActTimeDeviation( $G$ )

$\text{SemClaims}_{\mathcal{B}}(G, P)$  and  $\text{red}_{P,\tau}(E_\tau(G)) \neq f_P$ .

ReuseDeviation( $G$ )

$\text{SemClaims}_{\mathcal{B}}(G, P)$  and  $\text{red}_{P,U}(E_U(G)) \neq \text{red}_{P,U}(\text{useEff}_P(T_B(P)))$ .

EmpiricalDeviation( $G$ )

$\text{SemClaims}_{\mathcal{B}}(G, P)$  and the transported empirical cells are not equal to  $e_P$  in the fixed  $O_P$  fiber.

ComparisonDeviation( $G$ )

The empirical cells represent  $e_P$  but  $\text{cmp}_{E_T(G)}|_{W_P^{\text{act}}} \neq \omega_P|_{W_P^{\text{act}}}$ .

StandingTokenDeviation( $G$ )

The empirical and active comparison coordinates agree with  $e_P$  and  $\omega_P$ , but  $\text{st}_{E_T(G)}|_{Z_P^{\text{act}}} \neq \sigma_P|_{Z_P^{\text{act}}}$ .

BellLocalCompletion( $G$ )

$\Gamma(\text{emp}(E_T(G))) \neq \emptyset$ .

StandingFlow( $G$ )

There are  $G_0$  and a kernel-licensed  $\Phi : G_0 \rightarrow G$  with  $\text{SemClaims}_{\mathcal{B}}(G_0, P)$  such that one of

$\text{CreatesStand}_{\mathcal{B}}(\Phi)$ ,  $\text{AmplifiesStand}_{\mathcal{B}}(\Phi)$ ,  $\text{TransfersStand}_{\mathcal{B}}(\Phi)$ ,  $\text{MetricAuthorizes}_{\mathcal{B}}(\Phi)$

holds.

SameActRepair( $G$ )

There are  $G_0$  and a kernel-licensed  $\Phi : G_0 \rightarrow G$  such that  $\text{SemClaims}_{\mathcal{B}}(G_0, P)$  and  $\text{RepairsSameAct}_{\mathcal{B}}(\Phi)$ .

ScopeChange( $G$ )

There is no  $h : O_G \rightarrow O_P$  in  $\mathcal{G}_{\mathcal{B}}$  carrying the target identifier, cover, channels, and act identifier to those of  $P$ .

**Theorem 4.38** (Active-coordinate exhaustion). *Let  $G \in \text{BellGov}_{\mathcal{B}}$  satisfy  $\text{SemClaims}_{\mathcal{B}}(G, P)$ . If its*

target-active profile differs from the canonical active profile, then at least one of the following holds:

$$\begin{aligned} & \text{AnchorDeviation}(G), \quad \text{ContinuationDeviation}(G), \\ & \text{ActTimeDeviation}(G), \quad \text{ReuseDeviation}(G), \\ & \text{EmpiricalDeviation}(G), \quad \text{ComparisonDeviation}(G), \\ & \text{StandingTokenDeviation}(G), \quad \text{BellLocalCompletion}(G), \\ & \text{StandingFlow}(G), \quad \text{SameActRepair}(G), \\ & \text{ScopeChange}(G). \end{aligned}$$

*This theorem is a typed coordinate exhaustion only; it does not use later componentwise uniqueness to close continuation, temporal, or reuse deviations.*

*Proof.* Transport to the  $O_P$  fiber and compare the six active reductions. A domain, anchor, continuation, status, or use difference gives the corresponding first-line deviation. If those agree, the active standing effect has exactly three blocks: empirical cells, comparison values, and active standing tokens. A difference gives empirical, comparison, or standing-token deviation. Bell-local empirical replacement is the distinguished global-completion subcase. A difference produced by a same-domain modification is additionally classified as standing flow or repair. These cases exhaust inequality of the reduced tuple.  $\square$

**Definition 4.39** (Joint standing class). A joint standing class is a Bell standing class whose exact standing object satisfies the compatibility certificate. It is Bell-nonlocal when  $\Gamma(e_P) = \emptyset$  and quantum-entangled when it additionally has an entangled realization in the carrier of theorem 11.1.

**Theorem 4.40** (Joint standing class as the Bell-nonlocal standing form). *A compatible Bell-nonlocal empirical model used as one non-degenerate witness has the same-domain standing form of a joint standing class carrying  $T_B(P)$ . It is not representable on that domain as independently sufficient local standing plus a later connector.*

*Proof.* The standing quotient fixes the witness profile, theorem 4.33 certifies  $T_B(P)$ , and theorem 4.36 gives exact same-domain uniqueness. Independently sufficient local standing supplies a global section. A later connector instantiates standing transfer, a second invariant, or domain change.  $\square$

**Relation to established frameworks.** The global-section construction makes the relation to Fine and Abramsky–Brandenburger explicit:  $\text{JOB}_{\mathcal{B}}(P)$  carries the same empirical model and obstruction to Bell-local global completion, while AASC adds fixed-domain anchor, act-time, and reuse roles. Jarrett–Shimony locality distinctions concern factorization inside candidate causal or ontological models; the present theorem classifies the reusable witness and does not infer the absence of every richer-scope nonlocal model. Spekkens-style operational equivalence and extensional governance equivalence are neighboring but distinct structures [6, 7, 9].

## 5 AMetric co-standing and exact local realization

Quantum theory or another operational carrier supplies the numerical context distributions. AASC fixes the admissible same-domain type of the object carrying them. The Bell wings are local roles of one compatible empirical model, not independently complete standing states that acquire the fixed witness by later communication.

**Definition 5.1** (Local Bell face). For  $T_B(P)$  and setting  $x$ , the  $A$ -local face is the restriction of  $\mu_{xy}^P$  to  $A_x$  through its declared channel; the result is independent of  $y$  by overlap consistency. The  $B$ -local face is defined analogously. A local face is a restriction of the joint empirical model, not an independently sufficient local standing state.

**Definition 5.2** (AMetric co-standing). The two wings are in AMetric co-standing iff their standing-bearing compatibility is admitted as one joint object before metric distance, temporal order, propagation speed, or signal transport can function as admission-level discriminators. “Before” denotes priority in admissibility, not temporal precedence.

**Definition 5.3** (Boundary co-fixation). Boundary co-fixation is joint admission of the trace-fixed interfaces and  $T_B(P)$ . It fixes what local measurements sample. It does not assign pre-existing values to all settings, transmit information, or repair a distant standing state.

**Definition 5.4** (Exact realization family). An exact realization family for the fixed Bell profile is a family of normalized context measures  $\{\nu_{xy}\}$  having the admitted marginals, canonical covariance, and

$$\nu_{xy}(a, b) = P(a, b \mid x, y)$$

for every context and outcome pair.

**Theorem 5.5** (Realization universal property). *The exact context family  $\{\mu_{xy}^P\}$  is the unique exact realization family up to canonical isomorphism. It determines  $T_B(P)$  through theorem 4.34, and every object satisfying the exact compatibility certificate recovers that same family.*

*Proof.* Exactness fixes every context measure. Canonical covariance allows only lawful relabeling. The packet, restriction maps, and governance certificate in theorem 4.34 are then fixed by the witness profile and theorem 4.36.  $\square$

**Theorem 5.6** (AMetric co-standing carries the fixed Bell witness without communication typing). *The family  $\{\mu_{xy}^P\}$  carries the already supplied Bell correlation as contextwise faces of one boundary-cofixed compatibility object. No message, metric carrier, distant repair, or standing transfer is required or licensed as a same-domain component of that witness.*

*Proof.* The context realization certificate recovers the complete witness. The standing-flow theorem excludes creation, transport, and repair of standing as the same act. Boundary traces fix the local

interfaces without becoming outcome-conditioned selectors. The conclusion is a carrier classification; it does not derive the numerical probabilities from AMetric co-standing.  $\square$

| Transport reading                                                   | Boundary-compatibility reading                                                        |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Independent local standing states acquire correlation by influence. | Context distributions are local faces of one compatible empirical model.              |
| A local measurement completes the distant witness.                  | A local measurement samples a pre-admitted context distribution.                      |
| Metric separation demands a propagation account inside the witness. | Metric description is downstream of admission and does not supply standing authority. |
| Remote choice changes the local channel or standing status.         | Trace stability and operational no-signaling preserve local admission and marginals.  |

## 6 Measurement as admissible local realization

**Definition 6.1** (Preparation-level Bell compatibility). Let  $T_B^{\text{prep}}$  denote the compatibility object attached to the admitted preparation-level witness before the local acts that realize a particular context outcome. It is distinct from a post-measurement global state, a conditional state assignment, or the compatibility object of a later experiment.

**Definition 6.2** (Local measurement act). A local measurement act is an admissible same-domain act that samples the realization measure for one local context through a trace-fixed interface.

**Theorem 6.3** (Local measurement realizes pre-admitted Bell compatibility). *A local Bell measurement samples a local face of  $T_B^{\text{prep}}$  and produces a context outcome record. It does not retroactively create, transmit, amplify, or repair  $T_B^{\text{prep}}$  as the same witness act. The physical measurement may nevertheless change a post-measurement state, a conditional state assignment, later entanglement structure, or future correlation possibilities.*

*Proof.* The context distribution is already part of the admitted empirical model and is recovered by theorem 4.15. Retroactive creation or repair is excluded by theorem 3.20. Later physical or conditional states are distinct downstream objects and may change without altering the admission status of the earlier witness.  $\square$

**Proposition 6.4** (Local realization requires fixed witness-channel anchoring). *A local act samples  $\mu_{xy}^P$  only through an interface whose anchor status is fixed before the outcome. Choosing the interface after observing the result is post-selection or a new domain.*

*Proof.* The realization family is indexed by the declared setting/outcome channels. Boundary-trace factorization fixes their anchor status prior to witness use. A channel selected from the observed result is not the same fixed interface; it is post-selection, an independent selector, or a newly declared domain.  $\square$

**Proposition 6.5** (No delayed Bell calibration repair). *Delayed calibration, posterior detector identification, later coincidence-window choice, or retrospective relabeling cannot repair Bell classification as the same evaluated act. Such material is support for a later act, a declared update domain, or inadmissible same-act repair.*

*Proof.* Irreversibility fixes the status of the evaluated act at commitment. Later material may support a new analysis or declare a new channel package, but treating it as retroactively decisive changes the standing basis while claiming numerical identity of the act. That is the same-act repair excluded by the kernel.  $\square$

**Handoff to record determinacy.** The local act may produce a record occurrence. Whether that occurrence stands as the same retained evidential object is the target of the companion measurement-record theorem; the present paper governs the Bell compatibility sampled by the record.

## 7 Bell-local completion, hidden variables, and richer-scope models

The theorem excludes Bell-local completion on the fixed witness domain. It does not infer that every viable ontology must be local or that no declared richer-scope nonlocal model can reproduce the data.

**Definition 7.1** (Bell-local value determinacy). A Bell-local presentation consists of a setting-independent probability space  $(\Lambda, \Sigma, \mu)$  and normalized local stochastic kernels satisfying

$$P(a, b \mid x, y) = \int_{\Lambda} P_A(a \mid x, \lambda) P_B(b \mid y, \lambda) d\mu(\lambda).$$

The factorization includes parameter independence at the response level and conditional outcome independence given  $\lambda$ ; the operational no-signaling equalities follow after averaging when the setting-independence conditions hold [6, 7, 4].

**Proposition 7.2** (Compatibility determinacy is not local value determinacy). *The joint invariant fixes a compatible context family and its global-section obstruction. It does not assign one global value to every unperformed measurement and therefore is not a Bell-local hidden-variable completion.*

*Proof.* A Bell-local value completion provides a global section or an equivalent mixture of global assignments. For the Bell-nonlocal witness,  $\Gamma(e_P) = \emptyset$ . The compatibility object retains the context measures and their consistency without supplying the missing global assignment.  $\square$

**Theorem 7.3** (Bell-local factorization is unavailable on the fixed nonlocal witness). *If  $\Gamma(e_P) = \emptyset$ , no Bell-local factorization reproduces the fixed empirical model. A proposal that nevertheless adds*

*local authorizer variables must either fail extensionally, import additional standing-bearing structure, abandon setting independence, or declare a new domain.*

*Proof.* By theorem 4.12, any Bell-local factorization induces a global section. The empty global-section set therefore excludes extensional reproduction on the fixed domain. Altering setting independence or adding nonlocal, retrocausal, or superdeterministic structure changes the model class and must be declared as a wider domain.  $\square$

**Proposition 7.4** (Finite denominator-32 Bell-local certificate). *In the Lean companion, all sixteen deterministic Alice/Bob assignments yield scaled CHSH numerator 64 or  $-64$ . Every normalized denominator-32 convex mixture lies in the finite local envelope, while the vector/Born table has numerator 80. Thus no model in the formalized denominator-32 Bell-local class matches the witness.*

*Proof.* The direct Lean theorems enumerate the sixteen deterministic assignments, classify their extrema, enumerate the denominator-32 convex envelope, and prove that numerator 80 is absent from it. The proposition reports that machine-checked finite certificate; it does not extend the exclusion beyond the formalized model class.  $\square$

**Corollary 7.5** (Richer-scope hidden variables). *A nonlocal, superdeterministic, retrocausal, or otherwise enriched hidden-variable model is not excluded merely by the fixed witness. Its additional causal or ontic structure must be declared and assessed as a wider domain. Causal-model analyses show that such explanations face their own independence and fine-tuning burdens, but those are not collapsed into the present same-domain theorem [10, 11].*

*Proof.* The fixed-domain exclusion targets Bell-local completion and undeclared same-domain authority. Each listed model changes at least the causal, ontic, or setting-independence structure. It is therefore outside the original model class and must be judged on the declared enriched domain rather than treated as a same-domain counterexample.  $\square$

## 8 No-signaling hierarchy and same-domain transport exclusion

**Definition 8.1** (No-signaling hierarchy). For the fixed Bell domain:

- (NS1) **Operational marginal invariance** (NS<sub>1</sub>): remote settings do not change the local outcome marginals.
- (NS2) **No controllable communication channel** (NS<sub>2</sub>): no admissible local choice controllably modulates the remote marginal to encode a message.
- (NS3) **No same-domain standing-transport typing** (NS<sub>3</sub>): the fixed witness does not carry metric transport, distant repair, or standing transfer as one of its governance-bearing components.

In the general theorem,  $\text{NS}_1$  is compatibility on overlaps of the empirical model. In the finite Lean carrier, it is also derived from same/opposite symmetry. The passage from  $\text{NS}_1$  to  $\text{NS}_2$  is operational. The passage to  $\text{NS}_3$  uses the primitive no-transfer theorem, AMetric boundary closure, kernel-activated semantic effect exhaustion, and same-scope operator closure. Because those closure results are established upstream in section 3, this use is acyclic.

**Lemma 8.2** (Wing-local trace invariance). *Remote setting change cannot alter local anchor admission while the local channel identity is preserved. Otherwise the remote setting would function as a second anchor classifier.*

*Proof.* Wing locality through  $\ell_P^C$  and fixed-fiber  $\text{Aut}_G(P)$ -stability are defining properties of the boundary trace. If a remote setting changed the local trace value without changing the declared local channel, local anchor admission would depend on an undeclared remote parameter and the remote setting would become a second same-domain classifier.  $\square$

**Theorem 8.3** (Operational no-signaling forbids a controllable communication channel).  *$\text{NS}_1$  implies  $\text{NS}_2$ : if the receiver's outcome distribution is invariant under the sender's setting choice, that choice cannot encode a controllable message in the receiver's local statistics.*

*Proof.* A communication protocol requires at least two sender choices that induce distinguishable receiver distributions. Under  $\text{NS}_1$ , every sender setting induces the same receiver marginal. No receiver-side statistical test can therefore decode the sender's choice from the local outcomes alone.  $\square$

**Theorem 8.4** (Primitive no-transfer yields the no-signaling determinacy endpoint). *Let  $e_P$  be a compatible Bell witness used as one non-degenerate object. Under the active kernel, boundary-trace fixation, the AMetric boundary, and theorem 3.20,  $\text{NS}_3$  holds. The witness does not license same-domain signal, metric transport, marginal modulation, distant repair, or standing transfer.*

*Proof.* Operational marginal modulation is absent by  $\text{NS}_1$ . Controllable communication is absent by theorem 8.3. Remote settings cannot change local anchor admission by theorem 8.2. Standing creation, transfer, and repair are excluded by theorem 3.20; the AMetric boundary excludes metric or propagation variables as admission-level authority. Finally, theorems 3.7 and 10.6 exhaust every rival same-domain transport proposal: it is standing transfer, a second invariant or selector, same-act repair, target/domain change, or governance-inert skin. Hence no governance-non-equivalent same-domain transport typing survives.  $\square$

**Corollary 8.5** (No superluminal communication as an operational corollary). *A Bell-nonlocal no-signaling witness cannot be used as a controllable superluminal communication channel. This does not by itself rule out every richer-scope nonlocal ontology.*

*Proof.* Apply theorem 8.3 to the no-signaling marginals of the empirical model. The conclusion is operational and makes no inference about undeclared ontological structure.  $\square$

**Scope of the endpoint.** The result is target-relative. It does not derive quantum no-signaling from the kernel, nor does it infer the metaphysical absence of all nonlocal processes from marginal invariance. It proves that the same fixed witness has no governance-bearing transport or repair component unless additional structure is declared.

## 9 Metric representation is downstream of Bell-witness fixation

The AMetric boundary is a statement about admissibility priority, not a hidden physical surface or an ultra-short distance scale. Metric coordinates, spacetime separation, and temporal order describe the realized experiment after the witness interfaces and compatibility object are fixed; they do not supply admission-level standing authority.

**Theorem 9.1** (Boundary-level fixation). *The compatibility object  $T_B(P)$  and its trace-fixed interfaces are admitted before metric distance, temporal order, or propagation speed become governance-bearing discriminators. Here “before” denotes order of admissibility, not earlier physical time.*

*Proof.* The anchor traces and compatibility object belong to the standing profile that defines the witness before any interior coordinate representation is used. The AMetric boundary theorem excludes metric variables from admission-level authority. Metric and temporal coordinates may describe the realized experiment downstream without altering that prior governance status.  $\square$

**Theorem 9.2** (Metric representation does not supply a transport component). *Representing the two local realizations at spacelike-separated coordinates does not add a metric-transport component to the fixed witness. The spacetime representation displays correlations already contained in the context distributions; it does not, by representation alone, create a signal, carrier, or standing transfer.*

*Proof.* The context distributions and anchor traces are fixed in the standing profile. Coordinate assignment is a lawful interior representation of the realized events. By theorem 3.20, admission-level standing cannot be created or transported by a same-domain act, and by theorem 2.8 lawful representation changes preserve the witness class. Thus metric separation is descriptive of the realization, not an additional governance-bearing mechanism.  $\square$

**Corollary 9.3** (The AMetric boundary is not a physical distance scale). *The AMetric boundary is not the Planck scale or any other metric cutoff. It is defined by the absence of metric authority at admission.*

*Proof.* A metric cutoff is still a value inside a metric structure. The AMetric boundary is defined by the absence of metric standing authority at admission, not by an extreme or minimal distance.  $\square$

## 10 CHSH audit and Bell-governance operators

### 10.1 Input and output presentations

The Hilbert-space presentation represents a quantum preparation by  $\rho_{AB}$  and local measurement operators. The operational presentation is  $P(a, b \mid x, y)$ , its marginals, and its comparison profile. The Born rule transports the Hilbert presentation to the operational witness. The general theorem below begins with the operational empirical model and does not assume a Hilbert-space carrier until the quantum specialization.

**Proposition 10.1** (CHSH incompatibility of local standing sufficiency). *In the binary scope, if the fixed witness were carried by independently sufficient local standing, then  $|\text{CHSH}| \leq 2$ . Hence  $|\text{CHSH}| > 2$  excludes that standing form.*

*Proof.* Independently sufficient local standing supplies Bell-local factorization and a global section by theorem 4.12. Fine’s theorem gives the complete CHSH bounds.  $\square$

### 10.2 Finite Lean mechanization of the Bell-side witness

**Scaled integer vector geometry.** The finite Lean carrier uses

$$A_0 = (1, 0), \quad A_1 = (0, 1), \quad B_0 = (-5, -5), \quad B_1 = (-5, 5),$$

with  $\|A_i\|^2 = 1$ ,  $\|B_j\|^2 = 50$ , and scaled products  $4A_i \cdot B_j$  reproducing the finite axis profile. This is an exact integer encoding of the finite witness geometry, not a claim that all four integer vectors are normalized physical axes.

**Vector/Born table.** For scaled correlation numerator  $c \in \{20, -20\}$ ,

$$w_{\text{same}}(c) = \frac{32 + c}{4}, \quad w_{\text{opp}}(c) = \frac{32 - c}{4}.$$

Thus positive-correlation contexts have same-outcome cell weight 13 and opposite-outcome cell weight 3, with the roles reversed for negative correlation. Every context sums to 32 and every local marginal sums to 16.

**Proposition 10.2** (Structural finite no-signaling). *For either local outcome,*

$$w_{\text{same}} + w_{\text{opp}} = 16,$$

*independent of the remote setting. The finite vector/Born table is therefore structurally no-signaling.*

**Proposition 10.3** (Exact finite CHSH separation). *The finite correlator numerators are 20, 20, 20, -20, so the CHSH numerator is 80. Every one of the sixteen deterministic local response*

tables has numerator 64 or  $-64$ ; every denominator-32 convex mixture lies in the finite local envelope from 64 to  $-64$  in steps of 4. Hence 80 lies outside the formalized denominator-32 Bell-local and hidden-variable class.

**General finite-denominator interface.** The released code also defines a general finite-denominator convex-local interface and places general local hidden-variable models inside it. The exact exclusion theorem in the release is denominator-32; no all-denominator exclusion is attributed to the current code.

**Bridge status.** The finite arithmetic does not by itself define AMetric co-standing or the full determinacy endpoint. Given the declared bridge structures

`VectorBornOperationalBridge` and `OperationalDynamicsBridge`,

Lean routes the verified operational package to the endpoint surfaces

`BoundaryLevelCompatibility` and `NoSignalingEndpoint`.

| Claim                         | Lean theorem                                                        | Status                |
|-------------------------------|---------------------------------------------------------------------|-----------------------|
| Normalization                 | <code>vectorBorn_table_normalized32</code>                          | Direct                |
| Alice structural no-signaling | <code>vectorBorn_no_signaling_from_structural_symmetry32</code>     | Direct                |
| Bob structural no-signaling   | <code>vectorBorn_bob_no_signaling_from_structural_symmetry32</code> | Direct                |
| CHSH numerator 80             | <code>vectorTableCHSH_value32</code>                                | Direct                |
| Deterministic extrema         | <code>deterministicAssignmentIndex_table_extreme32</code>           | Direct                |
| Convex-envelope exclusion     | <code>vectorCHSH_exceeds_convex_local_envelope32</code>             | Direct                |
| Denominator-32 LHV exclusion  | <code>vectorBorn_not_denominator32_local_hidden_variable32</code>   | Direct                |
| Finite flagship bundle        | <code>vectorBorn_flagship_spine</code>                              | Direct bundle         |
| Boundary compatibility        | <code>VectorBornBoundaryCompatibility</code>                        | Conditional bridge    |
| No-signaling endpoint         | <code>VectorBornNoSignalingEndpoint</code>                          | Conditional bridge    |
| Deepest released endpoint     | <code>reasonableDeepestFiniteEntanglement_spine</code>              | Bundled finite/bridge |

### 10.3 Semantic standing-relevant operators

**Definition 10.4** (Standing-relevant Bell modification). A standing-relevant Bell modification is a map  $\Phi : G \rightarrow G'$  in  $\text{BellGov}_B$  claimed to alter same-domain witness governance. Its semantic support is

$$\text{supp}_{\text{gov}}(\Phi) = \{L \in \{D, A, T, C, \tau, U\} : h_*^L(E_L(G)) \neq E_L(G')\},$$

where  $h$  is the lawful arrow carried by  $\Phi$ . A modification with empty support is skin or bookkeeping.

**Theorem 10.5** (Semantic Bell-operator exhaustion). *Every standing-relevant Bell modification has support in the six effect channels of theorem 2.23; after grouping standing content with licensed*

*continuation and reports with their licensed use, these are exactly the five same-scope governance loci of the upstream exhaustion theorem. There is no independent sixth same-domain governance route.*

*Proof.* The semantic effect profile was defined independently of the canonical tuple. The Bivalence Theorem exhausts same-scope governance work by five loci [17, Theorem 4.2]; ATS element-effect exhaustion and uniqueness of effect channel force every descriptive change through the corresponding role channel [18, Theorems 4C–4D]. The representation theorem transports that semantic classification into  $\mathfrak{C}_{\mathcal{B}}$  without introducing a new effect.  $\square$

**Theorem 10.6** (Closure of inadmissible operator support). *For the fixed non-degenerate Bell target, the following operator effects are unavailable on the same domain: an undeclared anchor selector, a second standing invariant, Bell-local value completion of a nonlocal witness, standing creation or transfer, same-act repair, metric admission authority, and target-domain alteration disguised as preservation. An operator changing no governance effect is skin.*

*Proof.* No second invariant and no kernel-neutral governance difference follow from the Bivalence and kernel rigidity theorems [17, 15]. Choice and totalization closure exclude hidden selection and completion operators [19]. Bell-local completion contradicts theorem 4.13. Standing-flow effects are excluded by theorem 3.20. A target-domain change is not same-domain by theorem 2.17.  $\square$

## 10.4 Canonical target representative and positive uniqueness

**Definition 10.7** (Target-induced canonical governance representative). Define

$$C_P^* := (O_P, d_P^*, \alpha_P^*, \text{stEff}_P(T_B(P)), \text{ctEff}_P(T_B(P)), \tau_P^*, \text{useEff}_P(T_B(P)), s_0) \in \mathfrak{C}_{\mathcal{B}}.$$

The six governance coordinates are separated:  $\text{stEff}_P$  contains only standing-bearing empirical and comparison content,  $\text{ctEff}_P$  contains realization and continuation, and  $\text{useEff}_P$  contains report/reuse behavior.

**Theorem 10.8** (Candidate admissibility of the target-induced representative).

$$\text{NDBellWitness}(P) \implies \text{Claims}_{\mathcal{B}}(C_P^*, P) \wedge \text{Adm}_{\mathcal{B}}^{\text{cand}}(C_P^*; P).$$

*Proof.* The canonical representative uses the target’s active predeclared interface, empirical model, positive and negative evaluated standing obligations, exact realization support, mandatory-generated continuation, final status class, and mandatory supported report/reuse practice. Candidate admissibility contains no incoming-map or no-flow clause. The independent standing-flow theorem is used only downstream to exclude kernel-licensed same-act creation, transfer, repair, and metric authority. None of these checks presupposes the later uniqueness theorem.  $\square$

**Proposition 10.9** (Domain-effect uniqueness). *If  $\text{Claims}_{\mathcal{B}}(C, P)$  and  $\text{Adm}_{\mathcal{B}}^{\text{cand}}(C; P)$ , then  $E_D(C)$  and  $E_D(C_P^*)$  occupy the same lawful orbit.*

*Proof.* This is the fixed-target part of  $\text{Claims}_{\mathcal{B}}(C, P)$ . A different orbit is scope change.  $\square$

**Lemma 10.10** (Active standing-classification uniqueness). *If  $\text{StandCand}_P(t)$  and  $\text{UseAdeq}_P(u; t, c)$ , then*

$$\text{st}_t|_{\mathbf{Z}_P^{\text{act}}} = \sigma_P|_{\mathbf{Z}_P^{\text{act}}}.$$

*Proof.* Standing candidacy fixes  $z_P$  positive and every mandatory negative token to zero. Every other token in  $\mathbf{Z}_P^+$  supports a mandatory report or licensed reuse state, so mandatory-use adequacy forces it positive. Hence the active valuation is  $\sigma_P^{\text{act}}$ , which target adequacy identifies with  $\sigma_P$  on the active fragment. Coordinates outside that fragment have no target-bearing use and are skin.  $\square$

**Proposition 10.11** (Active anchor-effect uniqueness). *If  $\text{Claims}_{\mathcal{B}}(C, P)$  and  $\text{Adm}_{\mathcal{B}}^{\text{cand}}(C; P)$ , then*

$$\text{red}_{P,A}(E_A(C)) = \alpha_P^*|_{\mathcal{C}_P^{\text{act}}} = \text{red}_{P,A}(E_A(C_P^*))$$

*after lawful transport.*

*Proof.* This is the target-active equality in  $\text{TraceAdeq}_P(E_A(C))$ . The equality is an identity condition of the mandatory interface; the nontrivial closure result is that all richer anchor structure factors through a predeclared trace and cannot become a second classifier. A difference outside  $\mathcal{C}_P^{\text{act}}$  does not affect the witness practice and is skin.  $\square$

**Proposition 10.12** (Active standing-content uniqueness). *If  $\text{Claims}_{\mathcal{B}}(C, P)$  and  $\text{Adm}_{\mathcal{B}}^{\text{cand}}(C; P)$ , then*

$$\text{red}_{P,T}(E_T(C)) = \text{red}_{P,T}(\text{stEff}_P(T_B(P))) = \text{red}_{P,T}(E_T(C_P^*))$$

*after lawful transport.*

*Proof.* The target claim supplies a Bell-lawful transport into the  $O_P$  fiber for which the empirical cells equal  $P$ . Comparison coordinates are functions of those cells. The reviewed token and mandatory negative obligations are fixed by  $\text{StandCand}_P$ ; every remaining active positive token supports a mandatory report or reuse act and is positive by  $\text{UseAdeq}_P$ . Thus the active standing valuation is  $\sigma_P^{\text{act}}$ . These are exactly the coordinates retained by  $\text{red}_{P,T}$ .  $\square$

**Proposition 10.13** (Active realization and continuation uniqueness). *Suppose  $\text{Claims}_{\mathcal{B}}(C, P)$  and  $\text{Adm}_{\mathcal{B}}^{\text{cand}}(C; P)$ . Then*

$$\text{red}_{P,C}(E_C(C)) = \text{red}_{P,C}(\text{ctEff}_P(T_B(P))) = \text{red}_{P,C}(E_C(C_P^*))$$

*after lawful transport.*

*Proof.*  $\text{ContCand}_P$  makes joint and one-wing realization equal to the support and marginals of the transported empirical cells, hence to those of  $P$ . Mandatory generation and same-scope closure give  $R_C \equiv_{\text{cont},P} \mathcal{K}_P^{\text{act}}$  by theorems 3.13 and 3.21. The target-active continuation reduction records the  $\equiv_{\text{cont},P}$ -class, so the realization and target-bearing continuation coordinates agree. Governance-inert edge differences are skin.  $\square$

**Proposition 10.14** (Act-time class uniqueness). *If  $\text{Claims}_{\mathcal{B}}(C, P)$  and  $\text{Adm}_{\mathcal{B}}^{\text{cand}}(C; P)$ , then*

$$\text{red}_{P,\tau}(E_{\tau}(C)) = f_P = \text{red}_{P,\tau}(E_{\tau}(C_P^*)).$$

*Proof.* This is  $\text{TimeAdeq}_P$ . Distinct raw status labels with the same target-bearing final consequences occupy one status class and are governance-equivalent; a different class is nonfinal, repair, or a fresh act.  $\square$

**Proposition 10.15** (Mandatory report and reuse uniqueness). *If  $\text{Claims}_{\mathcal{B}}(C, P)$  and  $\text{Adm}_{\mathcal{B}}^{\text{cand}}(C; P)$ , then*

$$\text{red}_{P,U}(E_U(C)) = \text{red}_{P,U}(\text{useEff}_P(T_B(P))) = \text{red}_{P,U}(E_U(C_P^*))$$

*after lawful transport.*

*Proof.*  $\text{UseAdeq}_P$  makes every mandatory active act positive and requires standing support. The non-degenerate target licenses the same mandatory acts. Thus the restrictions to  $\mathcal{U}_P^{\text{act}}$  agree. Optional and inactive report tokens do not alter the target practice and are skin.  $\square$

**Theorem 10.16** (Candidate admissibility forces active exact realization). *For every  $C \in \mathfrak{C}_{\mathcal{B}}$ ,*

$$\text{Claims}_{\mathcal{B}}(C, P) \wedge \text{Adm}_{\mathcal{B}}^{\text{cand}}(C; P) \implies \text{Realizes}_{\mathcal{B}}(C, P).$$

*Proof.* Domain-orbit equality is theorem 10.9; active anchor, standing, continuation, status, and use equality are theorems 10.11 to 10.15. By theorem 3.10, these are exactly the active target equations.  $\square$

**Theorem 10.17** (Boundary-compatible governance-class collapse). *For every  $C \in \mathfrak{C}_{\mathcal{B}}$ ,*

$$\text{Claims}_{\mathcal{B}}(C, P) \wedge \text{Adm}_{\mathcal{B}}^{\text{cand}}(C; P) \implies C \equiv_{\text{gov}} C_P^*.$$

*Proof.* Candidate admissibility forces active exact realization by theorem 10.16. Target-active exact realization is extensional governance equivalence by theorem 2.23. Internal and inactive differences changing no active behavior are skin by theorem 3.7. Standing-flow exclusion is a separate consequence of theorem 3.20, not a premise of construction occupancy.  $\square$

**Corollary 10.18** (Full role-irreducibility after component closure). *If  $G \in \text{BellGov}_{\mathcal{B}}$  claims  $P$  and  $E_B(G)$  is candidate-admissible, every deviation in theorem 4.38 is closed and*

$$E_B(G) \equiv_{\text{gov}} C_P^*.$$

*Proof.* Coordinate-deviation exhaustion is theorem 4.38. Scope, flow, repair, Bell-local completion, empirical, comparison, and active-standing deviations are closed by the target, no-flow, Bell-obstruction, and standing-adequacy theorems. Anchor, continuation, status, and use deviations are closed by the componentwise propositions above.  $\square$

## 11 Bell-witness theorem and finite-dimensional quantum specialization

**Definition 11.1** (Finite-dimensional quantum realization). A finite-dimensional quantum realization of  $P$  in the specified bipartite tensor-product and local-POVM carrier is a density operator  $\rho_{AB}$  on finite-dimensional  $\mathcal{H}_A \otimes \mathcal{H}_B$  and local POVMs  $\{M_{a|x}^A\}$  and  $\{M_{b|y}^B\}$  such that

$$P(a, b \mid x, y) = \text{Tr}[\rho_{AB}(M_{a|x}^A \otimes M_{b|y}^B)].$$

The realization is separable iff

$$\rho_{AB} = \sum_{k=1}^n q_k \rho_k^A \otimes \rho_k^B, \quad q_k \geq 0, \quad \sum_k q_k = 1.$$

**Theorem 11.2** (Finite-dimensional separable realizations are Bell-local). *If a finite-dimensional realization of  $P$  is separable, then every operational family generated by its local POVMs has a Bell-local factorization and a global section.*

*Proof.* For the decomposition in theorem 11.1,

$$P(a, b \mid x, y) = \sum_k q_k \text{Tr}(\rho_k^A M_{a|x}^A) \text{Tr}(\rho_k^B M_{b|y}^B),$$

which is Bell-local factorization with hidden variable  $k$ . By theorem 4.12, it has a global section.  $\square$

**Theorem 11.3** (Boundary-compatible standing-form theorem). *Let  $P$  be a finite compatible Bell-nonlocal empirical model instantiated as one non-degenerate reusable witness on the fixed Bell domain. Then:*

- (i) *target instantiation activates the kernel roles and occupies a candidate-admissible governance realization without invoking standing-flow exclusion or later uniqueness;*

- (ii) every same-target typed AASC-facing Bell-governance presentation admits an equivariant encoding in the canonical effect carrier;
- (iii)  $T_B(P) \in \text{BellCompat}_{\mathcal{B}}(P)$  is certified and every exact-faithful compatibility object is  $\equiv_{\text{comp}}$ -equivalent to it;
- (iv) every same-target candidate-admissible governance construction has the same target-active anchor, standing, realization/continuation, final-status, mandatory report/reuse, and domain behavior as  $C_P^*$ ;
- (v) therefore every such construction is  $\equiv_{\text{gov}}$ -equivalent to  $C_P^*$ , while inactive differences are skin;
- (vi) operational no-signaling excludes controllable communication, and primitive standing-flow exclusion removes same-domain standing creation, amplification, cross-wing transfer, distant repair, and metric admission authority.

Thus boundary-level compatibility is the unique admissible target-bearing role form of the fixed Bell-nonlocal witness.

*Proof.* Occupancy and kernel activation are theorems 3.4, 3.6 and 3.15. Typed-interface representation and equivariance are theorems 2.28, 3.8 and 3.9. Compatibility certification and exact-faithful uniqueness are theorems 4.33 and 4.36. Coordinate exhaustion is theorem 4.38; target-active componentwise closure and class collapse are theorems 10.9, 10.11 to 10.15, 10.17 and 10.18. The no-signaling conclusions are theorems 8.3 and 8.4.  $\square$

**Proposition 11.4** (Governance-quotient rigidity). *If  $C_1, C_2 \in \mathfrak{C}_{\mathcal{B}}$  both claim  $P$  and satisfy  $\text{Adm}_{\mathcal{B}}^{\text{cand}}(C_i; P)$ , then*

$$C_1 \equiv_{\text{gov}} C_2.$$

*Proof.* By theorem 10.17, each is governance-equivalent to  $C_P^*$ .  $\square$

**Corollary 11.5** (Singleton target-induced governance quotient). *Under the hypotheses of theorem 11.3,*

$$\mathcal{Q}_P = \{[C]_{\equiv_{\text{gov}}} : \text{Claims}_{\mathcal{B}}(C, P) \wedge \text{Adm}_{\mathcal{B}}^{\text{cand}}(C; P)\}$$

*has exactly one element.*

*Proof.* Target occupancy supplies at least one candidate-admissible class. Candidate-admissibility rigidity supplies at most one. Therefore the quotient is a singleton.  $\square$

**Corollary 11.6** (Bell-nonlocal quantum entanglement as boundary-level compatibility). *Suppose  $P$  satisfies theorem 11.3 and has a finite-dimensional realization in the bipartite tensor-product and local-POVM carrier of theorem 11.1. Then every state realizing  $P$  within that carrier is entangled, and its Bell-witness role is the boundary-level compatibility form  $T_B(P)$ .*

*Proof.* A separable realizing state would be Bell-local by theorem 11.2, contradicting Bell nonlocality. The compatibility conclusion is theorem 11.3.  $\square$

**Corollary 11.7** (Post-quantum scope boundary). *A Bell-nonlocal no-signaling empirical model outside the quantum set satisfies the general witness theorem whenever it is instantiated as a non-degenerate reusable witness, but it has no quantum-entanglement corollary unless a finite-dimensional realization of theorem 11.1 is supplied.*

## 12 Mechanism constraint and paired handoff

The theorem does not choose a constructive mechanism for Bell-correlation generation. It constrains the governance-bearing content of any account that claims to preserve the same reusable Bell witness once that claim is supplied through the faithful typed Bell-governance interface. The repository module name **DerivedDynamics** denotes the finite operational derivation layer. It does not introduce a constructive physical law generating entanglement, a collapse dynamics, or a transport mechanism. A candidate mechanism, interpretation, or reconstruction must preserve Bell-witness identity, the target-active declared interface, no-signaling marginals, tensor-level compatibility, lawful redescription, the final status class, mandatory supported reuse, and the absence of hidden selector, delayed repair, and scope transport. If it does not, it is not a same-domain account of the Bell witness.

**Proposition 12.1** (Mechanism-to-role-space discipline). *Any proposed physical, dynamical, ontic, environmental, detector-channel, boundary, Everettian, or interpretive mechanism equipped with a faithful same-target AASC-facing Bell-governance interface is a semantic Bell-governance presentation and is classified by theorems 2.23 and 10.5. The theorem is universal over that typed interface. A native formalism not yet equipped with the interface enters the theorem only after its target-bearing effects are extracted and verified. If the interfaced proposal preserves the same non-degenerate witness, its standing-bearing work factors through the boundary-compatible normal form; additional standing-bearing structure must be declared as a new scope.*

*Proof.* Once the faithful interface is supplied, a proposed mechanism matters to the theorem only through its anchor, tensor/quotient, skin, temporal/status, or domain role. The semantic operator theorem classifies each contribution, and theorem 10.17 identifies the single governance class preserving the fixed target without hidden selector authority, same-act repair, or standing transfer. Interface extraction from a new native formalism is a separate representation task, not an unspoken premise.  $\square$

**Record determinacy and Bell-witness determinacy.**

| Measurement-record theorem                        | Bell-witness theorem                                              |
|---------------------------------------------------|-------------------------------------------------------------------|
| Record-bearing construction token $d$ .           | Bell witness table $P(a, b \mid x, y)$ .                          |
| Physical record occurrence.                       | Physical Bell correlation occurrence.                             |
| Ontic non-degenerate record determinacy.          | Ontic non-degenerate Bell-witness determinacy.                    |
| Record-role fixation.                             | Witness-channel and compatibility fixation.                       |
| Measurement-record quotient $ARef/\approx_{MR}$ . | Bell standing class $[p]_{\sim_B}$ .                              |
| Boundary trace for record role.                   | Boundary trace for setting/outcome/detector channel.              |
| Record standing.                                  | Witness standing.                                                 |
| Assignment-relative record failure.               | Hidden-load, repair, non-trace selector, or scope-change failure. |
| Boundary-trace normal form.                       | Boundary-level tensor-compatibility normal form.                  |

The record paper governs how measurement outcomes can stand as evidential records. The present paper governs how the witness structure sampled by such records can stand as a reusable Bell object. A later constructive dynamics, collapse theory, hidden-variable enrichment, Everettian restriction, or physical selection account must satisfy both: record determinacy must be fixed, and Bell-witness compatibility must not be retyped as hidden values, signaling, repair, or undeclared mechanism.

## 13 Scope, nonclaims, and denial burden

### 13.1 What the theorem establishes

The manuscript establishes:

1. target-instantiation occupancy and kernel necessity for non-degenerate Bell-witness use;
2. a lawful operational groupoid together with an independent empirical lawful groupoid preserving global-section status;
3. a typed AASC-facing Bell-governance interface with six explicit effect spaces and coherence data;
4. an equivariant representation theorem and orbit-valued canonical normalization;
5. Bell-profile completeness and its standing-quotient corollary;
6. Bell-local/global-section equivalence and the Bell joint obstruction packet;
7. a raw compatibility carrier, an explicit certificate predicate, the certified class  $\text{BellCompat}_B(P)$ , compatibility equivalence  $\equiv_{\text{comp}}$ , a typed standing-effect map, and a canonical embedding into the governance carrier;
8. certification and exact compatibility uniqueness for  $T_B(P)$ ;

9. coordinate-deviation exhaustion over every same-target typed Bell-governance presentation, followed by completed role collapse;
10. explicit same-domain standing-flow exclusion;
11. positive derivation of anchor, compatibility, realization, act-time, reuse, and domain behavior from primitive candidate admissibility;
12. the boundary-compatible standing-form theorem as the principal endpoint;
13. singleton uniqueness in the target-induced governance quotient as its occupancy-and-rigidity corollary; and
14. a general Bell-nonlocal no-signaling theorem with a finite-dimensional, carrier-qualified entanglement specialization.

### 13.2 What the theorem does not claim

The manuscript does not claim:

1. a first-principles derivation of Hilbert-space quantum mechanics, the continuum Born rule, Tsirelson's bound, or experimental Bell violation;
2. a constructive mechanism generating Bell correlations;
3. that every entangled state is Bell-nonlocal;
4. that every Bell-nonlocal no-signaling model is quantum-realizable;
5. exclusion of declared richer-scope nonlocal, superdeterministic, retrocausal, or causal models;
6. coordinate minimality of  $\text{JOb}_{\mathcal{B}}(P)$ ; the proved irreducibility is governance-role irreducibility;
7. an all-denominator Lean exclusion theorem; the released exact finite exclusion is denominator-32; or
8. that the released Lean package formalizes the v0.19 correctness-closure theorem layer.

### 13.3 Standing-filtered denial matrix

A same-mode refutation must do at least one of the following:

- (D1) deny target instantiation or break target-instantiation occupancy;
- (D2) break vocabulary-neutral adequacy to kernel necessity;
- (D3) break an exact imported theorem or show that its stated Bell hypothesis fails;

- (D4) exhibit a faithful operational equivalence not represented in  $\mathcal{G}_B$  and carrying a new governance effect;
- (D5) construct a typed Bell-governance interface whose effects do not factor through the six effect spaces;
- (D6) break equivariance of the semantic encoding or normalization;
- (D7) exhibit an independent same-domain governance effect outside the exact five-locus exhaustion;
- (D8) provide an exact faithful compatibility object not  $\equiv_{\text{comp}}$ -equivalent to  $T_B(P)$ ;
- (D9) defeat the Bell-local global-section obstruction;
- (D10) discharge the joint standing effect while avoiding every alternative in theorem 4.37;
- (D11) produce same-domain standing creation, amplification, transfer, repair, or metric admission authority without changing the governance profile;
- (D12) break candidate admissibility or one of the componentwise normal-form derivations; or
- (D13) supply a finite-dimensional separable realization of a Bell-nonlocal witness within the specified quantum carrier.

Different terminology, different internal data packaging, governance-inert metadata, target refusal, a richer declared domain, or a request for another proof class is not a same-mode refutation.

## 14 Conclusion

This paper proves a fixed-domain determinacy theorem for finite Bell-nonlocal no-signaling witnesses and a finite-dimensional Bell-entanglement specialization. The starting point is not optional adoption of AASC vocabulary. Once one operational family is used as the same non-degenerate witness, governed sameness and target occupancy are already in force. Reference, standing, admissibility, and irreversibility are the identity conditions under which the object remains determinate across lawful transformation, comparison, retained records, and reuse.

The proof is native-syntax-neutral but requires a mathematically typed AASC-facing governance interface. A Bell-governance presentation now consists of an internal carrier, a designated realization, six object-indexed effect sections, and an explicit coherence predicate. This interface admits causal models, instrument algebras, process structures, contextuality constructions, histories, categories, and other mathematical packages without treating their native syntax as a new governance role. The representation theorem encodes their induced effects in the canonical Bell carrier. Encoding is equivariant, while normalization is the invariant diagonal orbit of the target-active six-effect reduction and selects no privileged representative. Redundant ancillas, dilations, refinements,

coarse-grainings, duplicate labels, convex re-expressions, and other faithful implementation changes therefore do not create rival governance classes.

The Bell compatibility object is also fully typed. Its fields are the context measures, marginal restrictions, realization support, lawful continuation, covariance, Bell profile, and proof-bearing governance certificate. Compatibility equivalence is defined on these objects and connected to governance equivalence by a canonical embedding. The principal result is not the final cardinality statement by itself. It is the normal-form theorem: the weaker candidate-admissibility conditions force every same-target typed governance presentation into the certified compatibility effect  $T_B(P)$  and its associated realization, temporal, and reuse effects.

Coordinate-deviation exhaustion followed by the componentwise normal-form theorems closes the possible discharge routes. The joint standing work cannot be reduced to local marginals, reassigned to an independent anchor, erased as skin, completed by Bell-local values, transported between wings, repaired after the act, or preserved under a changed target. Operational no-signaling excludes controllable communication, while the standing-flow theorem independently excludes same-domain standing creation, amplification, transfer, distant repair, and metric admission authority. The correlation is therefore carried as lawful local realizations of one boundary-cofixed compatibility structure, not as a same-domain transport process.

Target occupancy supplies at least one candidate-admissible governance class. The Bell-specific componentwise theorems derive the target-active anchor, compatibility, realization/continuation, final-status class, mandatory report/reuse behavior, and domain effect from the primitive candidate conditions. The singleton governance quotient follows as the final occupancy-and-rigidity corollary. It is an identity and normal-form consequence of the already instantiated determinate object, not an independent postulate of existence.

When the empirical model has a finite-dimensional bipartite tensor-product realization with local POVMs, every separable state produces a Bell-local family. Bell nonlocality therefore forces every state realizing that witness within the specified carrier to be entangled. This is the title-level result: Bell-nonlocal quantum entanglement occupies the boundary-level compatibility role of the reusable witness. Post-quantum no-signaling models remain within the general witness theorem but outside the quantum corollary.

The finite Lean companion verifies the exact denominator-32 Bell carrier and its conditional endpoint routing. The typed semantic interface, empirical lawful groupoid, wing-indexed standing-flow structure, compatibility equivalence, coordinate exhaustion, completed role collapse, and candidate-to-canonical derivation, and finite-dimensional quantum specialization remain manuscript-level results outside the released Lean layer.

## A Paired quantum determinacy ledger

The measurement-record and Bell-witness papers share the same kernel necessity while retaining distinct target objects. The record paper fixes which physical record may stand as the same evidential object. The present paper fixes the standing form of the Bell witness sampled by such records. Neither theorem is a premise of the other.

| Shared constraint     | Measurement record                             | Bell witness                                         |
|-----------------------|------------------------------------------------|------------------------------------------------------|
| Non-degenerate target | Same evidential record object.                 | Same reusable empirical-model witness.               |
| Target occupancy      | Admissible record-role realization.            | Admissible Bell-governance realization.              |
| Standing object       | Measurement-record quotient.                   | Bell standing profile and compatibility object.      |
| Boundary trace        | Record-role trace.                             | Setting/outcome and channel trace.                   |
| Residual normal form  | Boundary-trace record fixation.                | Boundary-level compatibility.                        |
| Failure mode          | Assignment-relative or fail-closed record use. | Hidden load, repair, standing flow, or scope change. |

## B Bell-scope and mechanization certificate

For quick audit:

1. Non-degenerate same-witness use entails target occupancy and the kernel roles.
2. Every typed AASC-facing governance presentation has an equivariant canonical effect-equivalent Bell tuple.
3. Bell nonlocality is obstruction to a Bell-local global section.
4. The exact context family is normalized, overlap-consistent, covariant, and recovers the witness.
5.  $T_B(P)$  is a fully typed compatibility structure with explicit measure, restriction, realization, continuation, covariance, profile, and certificate fields.
6. The compatibility certificate is exact; every exact-faithful compatibility object is  $\equiv_{\text{comp}}$ -equivalent to  $T_B(P)$ .
7. Standing creation, amplification, transfer, metric admission authority, and same-act repair are excluded.
8. Every same-target candidate-admissible construction is forced to carry the canonical compatibility effect and is governance-equivalent to  $C_P^*$ .
9. The target-induced governance quotient has exactly one element.

10. In the finite Lean carrier, the vector/Born table has denominator 32, marginal numerator 16, CHSH numerator 80, deterministic local extrema  $\pm 64$ , and no denominator-32 local hidden-variable representation.
11. The deepest released Lean endpoint is `reasonableDeepestFiniteEntanglement_spine`; the endpoint bridge is conditional on declared bridge structures.

## C Boundary and non-claim guide

The AMetric boundary is an admissibility boundary, not a Planck-scale surface, hidden location, or physical transport medium. Boundary traces are anchor-effect representatives, not detector ontology.  $T_B(P)$  is a canonical governance object carrying the exact empirical model and obstruction profile; its irreducibility is role-theoretic rather than coordinate-minimal. Local measurement realizes a face of preparation-level compatibility and may still change a later conditional or post-measurement state. The theorem constrains same-domain witness typing; it does not choose a constructive dynamics.

## D Co-standing glossary

### Target-instantiation occupancy.

The canonical governance typing already occupied when a determinate witness exists as the target under review.

### Lawful Bell equivalence.

An isomorphism in the groupoid completion of exactly recoverable operational Bell simulations. It is represented by a finite zigzag and preserves the target data by construction.

### Typed Bell-governance interface.

An internal carrier with a designated realization, six typed governance-effect maps, and a coherence certificate for the fixed Bell target.

### Extensional governance equivalence.

Equality of target, anchor, standing, continuation, act-time, and reuse behavior, independent of internal representation.

### Compatibility equivalence.

Equality of context measures, restrictions, realization support, continuation, covariance, Bell profile, and report/reuse behavior for typed compatibility structures.

### Bell joint obstruction packet.

The canonical empirical-model orbit, fixed-marginal fiber, Bell-local subfiber, global-section status, and comparison profile.

**Role-irreducible joint content.**

Joint standing work that cannot be discharged into local marginals, anchors, skin, local values, transport, or repair.

**AMetric co-standing.**

Admission of the two wings as local roles of one compatibility object before metric structure may carry admission authority.

**Boundary co-fixation.**

Joint fixation of the trace-defined interfaces and the compatibility object sampled through them.

**Same-domain standing-flow exclusion.**

The prohibition on creating, amplifying, transferring, or repairing standing while preserving the same target and act.

**E Theorem-status map**

| Status                         | Representative result                                                                                                                                             | Role in the argument                                                          |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Standard Bell mathematics      | Bell-local factorization, global-section obstruction, separable-state locality                                                                                    | External carrier mathematics and quantum specialization.                      |
| Direct finite Lean theorem     | Denominator-32 normalization, structural no-signaling, CHSH 80, local envelope exclusion                                                                          | Machine-checked finite witness.                                               |
| Conditional Lean bridge        | Boundary and no-signaling endpoint surface from declared bridges                                                                                                  | Formal routing, not lower derivation of the manuscript theorem.               |
| Imported AASC theorem          | Kernel necessity, standing conservation, effect exhaustion, AMetric closure, trace fixation                                                                       | Upstream necessity and closure spine.                                         |
| Bell-specific specialization   | Standing-flow exclusion, Bell-profile completeness, object-indexed lawful effect functors                                                                         | Local verification of imported hypotheses and consequences.                   |
| New Bell-specific theorem      | Compatibility certification, empirical-lawful invariance, coordinate-deviation exhaustion, completed role collapse, candidate-to-canonical normal-form derivation | Principal manuscript contribution.                                            |
| Extensional/quotient corollary | Exact compatibility extensionality, Bell standing quotient, and singleton governance quotient                                                                     | Consequences of certification, completeness, occupancy, and derived rigidity. |

## F Lean 4 mechanization ledger

The released finite companion is archived under DOI 10.5281/zenodo.21287887 [24]. Its repository is

<https://github.com/somamaley-ux/quantum-entanglement-boundary-compatibility>.

Release v0.8.0 culminates in `DerivedDynamics.lean`. Its strongest named endpoint is

`reasonableDeepestFiniteEntanglement_spine`.

| Layer                  | Machine-checked content                                                                                                                                                                                                                                                                              | Claim status                               |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Finite arithmetic      | Integer vectors, scaled axis profile, vector/Born weights, normalization, marginals, and CHSH numerator 80.                                                                                                                                                                                          | Direct and unconditional.                  |
| Finite local envelope  | Sixteen deterministic tables, extrema $\pm 64$ , denominator-32 convex envelope, and exclusion of numerator 80.                                                                                                                                                                                      | Direct and unconditional.                  |
| AASC endpoint bridge   | Boundary trace, joint standing, tensor compatibility, AMetric co-standing, local realization faces, and endpoint exclusions.                                                                                                                                                                         | Conditional on declared bridge structures. |
| Manuscript v0.19 layer | Target occupancy, induced object-indexed effect functors, empirical lawful equivalence, wing-indexed standing-flow predicates, separated compatibility effects, coordinate exhaustion and completed role collapse, candidate-to-canonical derivation, and finite-dimensional quantum specialization. | Not in released Lean.                      |
| Outside the claim      | Hilbert reconstruction, continuum Born rule, Tsirelson’s bound, experimental derivation, and physical mechanism.                                                                                                                                                                                     | Not derived.                               |

## G Imported theorem and Bell-specialization ledger

The following table reproduces the load-bearing operative statements from the preferred source PDFs supplied with this revision. Page numbers are PDF pages. The project-level `IMPORTED_THEOREM_SPECIALIZATION_LEDGER.md` records the expanded statements and dependency map.

**Bell-side activation packet and local verification.** The imported conclusions are applied only after the primitive packet of theorem 3.5 has been established locally. In expanded form:

- (H1) one fixed non-degenerate Bell-witness target and one witness act: theorem 2.30;
- (H2) a fixed lawful orbit with object-indexed effect functors: theorems 2.9, 2.11 and 2.14;
- (H3) the declared verdict set  $\{0, 1\}$  on the evaluated fragment: theorems 2.1 and 3.1;

- (H4) one declared stable boundary-admission map: theorem 2.32;
- (H5) act-time finality: theorems 2.30 and 3.1;
- (H6) six syntactically typed query sorts and effect sections: theorems 2.3 and 2.20; and
- (H7) nontrivial comparison, continuation, report, and reuse data: theorems 2.1 and 2.14.

Neither no-second-classifier closure nor semantic effect exhaustiveness appears among H1–H7.

**Theorem G.1** (Bell activation of the imported fixed-domain interface). *Under H1–H7, the Bell target satisfies the primitive hypotheses used from the kernel, bivalence, anchor/tensor/skin, same-scope operator, AMetric-boundary, boundary-trace, and singleton-interior theorems listed below. Their Bell-specialized conclusions are exactly those stated in the third column.*

*Proof.* H1, H3, and H5 verify target determinacy, evaluability, and finality; H2 fixes the same carrier and lawful transformation class; H4 supplies one declared gate without assuming that no rival gate exists; H6 gives only a typed syntactic interface; and H7 supplies nontrivial standing-bearing use. Kernel necessity follows first. The imported bivalence, ATS, operator, AMetric, trace, and singleton-interior theorems then derive no second classifier, semantic effect exhaustion, unique effect-channel routing, standing conservation, and rigidity. Thus no imported conclusion is used as its own activation hypothesis.  $\square$

| Source node                         | Operative upstream statement                                                                                                                                              | Bell specialization                                                          |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Kernel, Lemma 2.6, p. 11 [15]       | Any target-adequate raw-sequence regime already requires reference, standing, irreversibility, and admissibility.                                                         | Neutral reusable-witness adequacy forces $K_B(P)$ .                          |
| Kernel, Definition 2.10, p. 13 [15] | Governance equivalence is sameness of governance work preserving target identity, act identity, status quotient, continuation, reportability, and redescription behavior. | Grounds extensional $\equiv_{\text{gov}}$ , not internal-data equality.      |
| Kernel, Theorem 2.34, p. 20 [15]    | No meaningful regime supports non-degenerate construction while lacking the full kernel or governance-equivalent replacements.                                            | Kernel denial loses or retypes the Bell target.                              |
| Kernel, Theorem 5.9, p. 28 [15]     | Any sufficient governance package must cover the same roles as the kernel; no governance-non-equivalent package survives.                                                 | Different Bell vocabularies may vary while governance work is fixed.         |
| Kernel, Theorem 6.14, p. 33 [15]    | The standing-instantiated admissible interior is extensionally unique.                                                                                                    | Any adequate Bell admissibility predicate agrees pointwise on standing acts. |
| Kernel, Theorem 6.15, p. 33 [15]    | No licensed same-domain operation creates, amplifies, transfers, or recovers standing.                                                                                    | Supplies the standing-flow exclusion.                                        |
| Kernel, Theorem 6.17, p. 34 [15]    | A candidate same-domain admissibility invariant is equivalent, illicit, a domain change, or bookkeeping.                                                                  | No rival Bell invariant with independent authority.                          |
| Kernel, Theorem 6.20, p. 36 [15]    | Target determinacy, step evaluability, act-time finality, and fidelity determine one governance-equivalence class of kernel structures.                                   | Fixed Bell adequacy activates one governance form.                           |

| Source node                                                            | Operative upstream statement                                                                                                             | Bell specialization                                                               |
|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Kernel, Corollary 10.4, p. 42 [15]                                     | Exactly one admissible interior exists up to governance equivalence.                                                                     | Supports singleton interior after target occupancy.                               |
| Bivalence, Theorem 3.4, p. 7 [17]                                      | A second invariant is bookkeeping or creates a second gate or repair channel; no second same-scope invariant exists.                     | Excludes a hidden Bell classifier beside the trace/kernel interface.              |
| Bivalence, Theorem 3.5, p. 8 [17]                                      | Every same-domain admissibility-preserving role is anchor, tensor, or skin; no fourth primitive role exists.                             | Underlies Bell effect routing.                                                    |
| Bivalence, Theorem 3.6, p. 8 [17]                                      | Same-domain governance cannot create, restore, transfer, or amplify standing authority.                                                  | Gives explicit Bell standing-flow closure.                                        |
| Bivalence, Theorem 4.2, p. 10 [17]                                     | Governance work is exhausted by five loci; there is no sixth same-domain route.                                                          | Semantic support theorem for arbitrary Bell presentations.                        |
| Bivalence, Theorems 5.3–5.4, pp. 12–13 [17]                            | Governance-relevant difference cannot be kernel-neutral; every same-domain package closes to the fixed governance form or changes scope. | Grounds representation and class-collapse results.                                |
| ATS, Theorem 4A, p. 15 [18]                                            | Every admissibility-bearing descriptive behavior descends through the fixed standing quotient.                                           | Bell-profile completeness.                                                        |
| ATS, Theorems 4C–4D, pp. 16–17 [18]                                    | Every element-level governance effect is exhausted and has a unique effect channel.                                                      | Arbitrary Bell formalism is classified by induced governance effect.              |
| ATS, Theorem 5 and Theorem 5A, pp. 19–21 [18]                          | Anchor, tensor, and skin are exhaustive, minimal, and role-disjoint.                                                                     | Canonical Bell role carrier.                                                      |
| ATS, Corollary 5A.2 and Theorem 5C, pp. 25–26 [18]                     | Null-effect metadata collapses to skin; role factorization is unique.                                                                    | Canonical normalization and extensional equivalence.                              |
| Operator closure, Lemma 2.10, p. 6 [19]                                | An auxiliary datum is already standing-fixed and bookkeeping, or imported and inadmissible.                                              | Covers hidden Bell selectors, evaluators, and completions.                        |
| Operator closure, Theorem 3.10 and Corollaries 3.11–3.12, pp. 8–9 [19] | No same-scope totalization; no post-hoc repair; no deferred admissibility.                                                               | Blocks delayed channel or witness completion.                                     |
| Operator closure, Theorem 4.5 and Result 5.2, pp. 10–11 [19]           | Totalization strategies are exhausted; every choice operator is inadmissible or bookkeeping-equivalent.                                  | Closes hidden selection and canonicalization authority.                           |
| AMetric, Theorem 2.5, p. 5 [21]                                        | No selector or parameter carries admissibility authority at the boundary.                                                                | Excludes metric, ranking, probability, and transport authority at Bell admission. |
| AMetric, Theorems 2.12–2.15, pp. 7–8 [21]                              | Boundary routes are exhausted and collapse into the fixed role structure.                                                                | No hidden Bell boundary route remains.                                            |
| AMetric, Theorem 5.13 and Definition 5.17, pp. 26–27 [21]              | The AMetric boundary is maximal pre-ontological structure and carries no metric selector.                                                | Supports AMetric co-standing without a physical surface.                          |
| Boundary trace, Theorem 3.5 and Corollary 3.6, p. 9 [22]               | Internal selection fails under constructional non-selectivity; after internal and indexical exhaustion, boundary coupling is forced.     | Motivates Bell interface trace fixation.                                          |
| Boundary trace, Definition 4.3 and Theorem 6.1, pp. 9 and 13 [22]      | A typed declared trace fixes the role anchor exactly in the singleton stable case.                                                       | Bell channel admission factors through declared traces.                           |
| Boundary trace, Theorems 7.2, 7.5–7.6, pp. 15–16 [22]                  | No environmental fixation survives without singleton stable coupling; symmetry-saturated alternatives are exhausted.                     | No detector/environment oracle or hidden selector.                                |
| Boundary trace, Proposition 10.6, p. 21 [22]                           | The environment itself is not the selector.                                                                                              | Environment may instantiate a trace, not supply independent authority.            |
| Unus, Theorem 6, p. 12 [20]                                            | In the standing-instantiated application regime, exactly one admissible interior exists.                                                 | Occupancy excludes the zero-class reading; rigidity excludes a second class.      |

| Source node                 | Operative upstream statement                                                    | Bell specialization                    |
|-----------------------------|---------------------------------------------------------------------------------|----------------------------------------|
| Unus, Theorem 7, p. 13 [20] | The apex closure theorem eliminates discrete rival interiors and repair routes. | Final Bell singleton-interior support. |

## H Reference integrity and publication notes

The external literature positions the theorem relative to Bell-local factorization, Fine’s global-distribution theorem, Jarrett–Shimony locality conditions, empirical-model/global-section contextuality, operational equivalence, causal-model fine-tuning, post-quantum no-signaling correlations, and the distinction between entanglement and Bell nonlocality. The preferred AASC source PDFs supplied with this revision were used to verify the theorem numbers, PDF pages, operative hypotheses, and Bell substitutions in section G. The Lean 4 development is a released finite companion. It verifies the finite witness and conditional routing surface; the typed v0.19 fixed-target governance and compatibility theorem layer remains manuscript-level. The theorem-status map in section E separates standard, direct Lean, conditional Lean, imported, specialized, and new results.

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